CHAPTER 08

# Carboxylic Acids

## Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: c204e0692096b7748c26d06c23423f7c1619e8e0b43ba3eeb456747621fd142b

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

### Coverage checklist

ch08:exercise:acid-ratio | acid-ratio:stage-1, acid-ratio:stage-2, acid-ratio:stage-3

ch08:exercise:make-ester | make-ester:stage-1, make-ester:stage-2, make-ester:stage-3

ch08:exercise:lose-co2 | lose-co2:stage-1, lose-co2:stage-2, lose-co2:stage-3

ch08:exercise:soap-head | soap-head:stage-1, soap-head:stage-2, soap-head:stage-3

ch08:lesson:acid-properties | acid-properties:rule, acid-properties:cue, acid-properties:trap, acid-properties:app, acid-properties:step-1, acid-properties:step-2

ch08:lesson:acid-reactions | acid-reactions:rule, acid-reactions:cue, acid-reactions:trap, acid-reactions:app, acid-reactions:step-1, acid-reactions:step-2, acid-reactions:step-3

ch08:lesson:reduction-decarboxylation | reduction-decarboxylation:rule, reduction-decarboxylation:cue, reduction-decarboxylation:trap, reduction-decarboxylation:app, reduction-decarboxylation:step-1, reduction-decarboxylation:step-2

ch08:lesson:soap | soap:rule, soap:cue, soap:trap, soap:app, soap:step-1, soap:step-2

ch08:molecule:aceticacid | aceticacid

ch08:molecule:acetate | acetate

ch08:molecule:ester | ester

ch08:molecule:acetamide | acetamide

ch08:molecule:acetoaceticacid | acetoaceticacid

ch08:molecule:acetone | acetone

ch08:molecule:laurate | laurate

ch08:curriculum | scope-1, scope-2

# Curriculum scope

scope-1: 8.1 Description and Properties

scope-2: 8.2 Reactions of Carboxylic Acids

Objective: Predict acid ionization and physical properties

Objective: Explain esterification, activation, and reduction

Objective: Track decarboxylation and saponification

Supplied Kaplan curriculum: Chapter 8; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter08b.xhtml

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

# acid-properties | Carboxylic acid structure and acidity

Two oxygens stabilize the conjugate base

## acid-properties:rule | Core Rules

A carboxyl group has C=O and OH bonded to the same carbon. Carboxylic acids hydrogen bond and can dimerize, especially in nonpolar media. Their carboxylates are resonance stabilized. Electron-withdrawing substituents generally increase acidity, with inductive effects weakening over distance.

## acid-properties:cue | Memory Cues

pH above pKa → more carboxylate. pH below pKa → more neutral acid.

## acid-properties:trap | Common Mistakes / MCAT Traps

The OH of a carboxyl group is not classified as an ordinary alcohol group. A ring can carry COOH; its carboxyl carbon need not belong to the ring. Larger hydrocarbon regions can still limit aqueous solubility.

## acid-properties:app | MCAT Applications

Ionization controls how fatty acids, drug acids, and amino-acid side chains partition between environments.

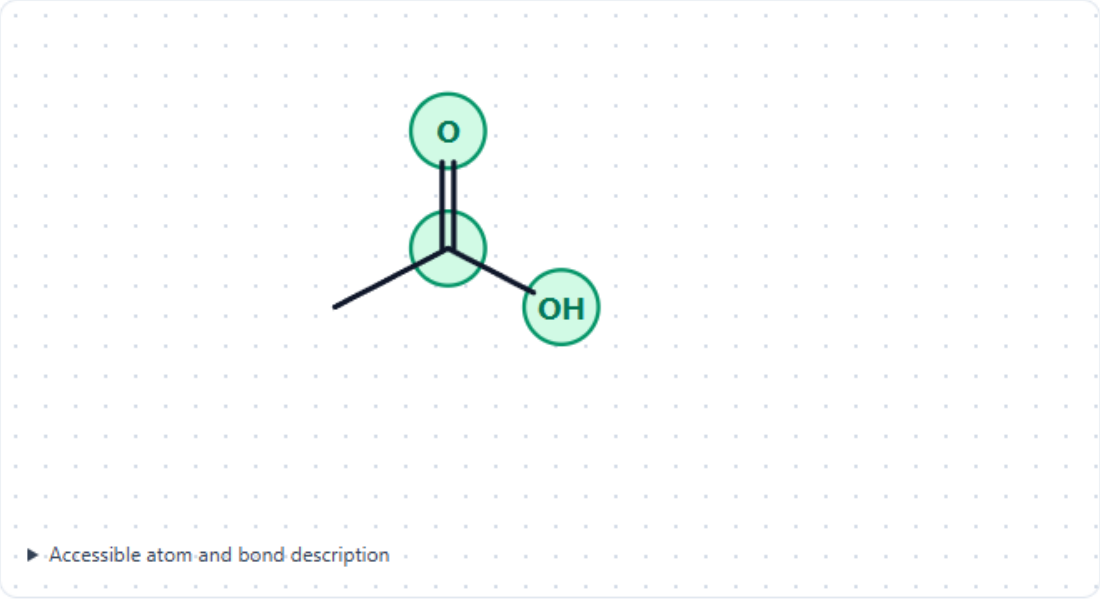
## acid-properties:step-1 | 1. Recognize COOH

The acid functional group includes both oxygens and their shared carbon. Its carbon has no room for another carbon substituent beyond the existing skeleton.

Displayed molecule ID: aceticacid

{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: acid-properties-step-1



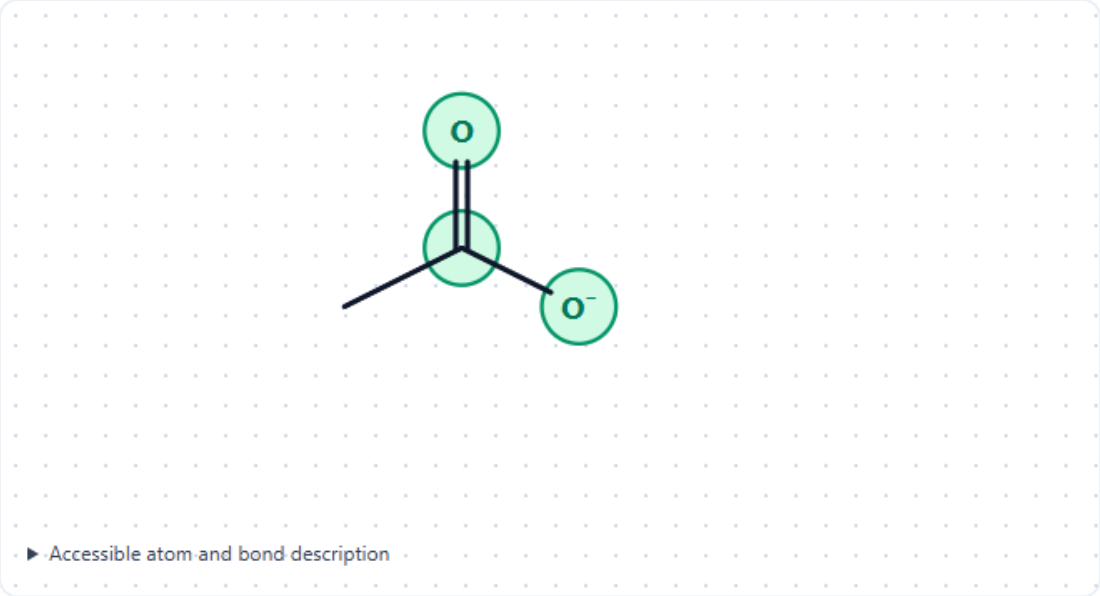

## acid-properties:step-2 | 2. Change pH

Adjust pH around a model pKa of 4.8. At pH=pKa, [A−]=[HA]; two units above pKa gives a 100:1 ratio, about 99% A−.

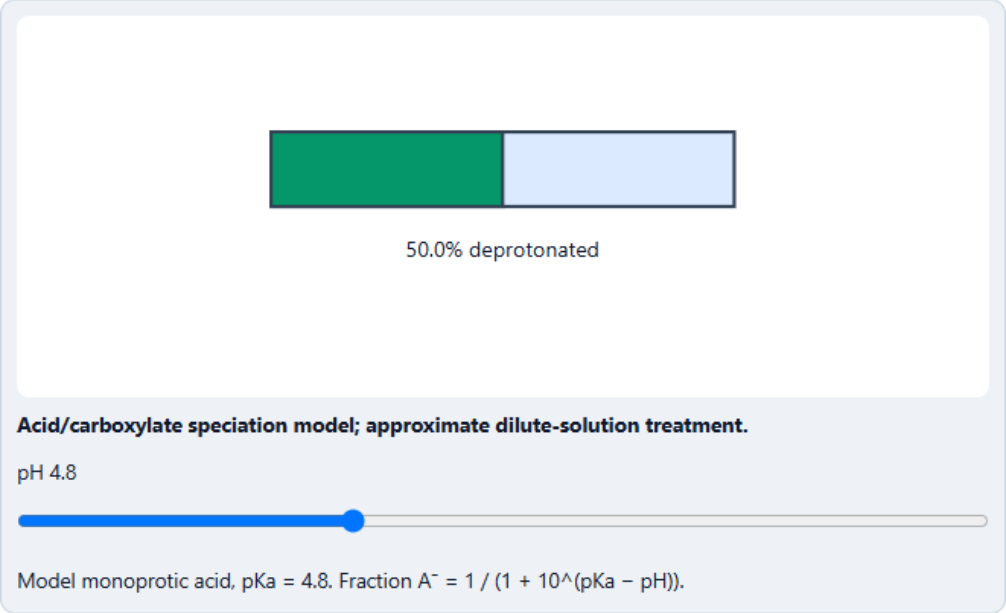
Displayed molecule ID: acetate

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Screenshot: acid-properties-step-2




Screenshot: acid-properties-step-2-diagram




{"diagramSpecification": {"type": "acid", "pKa": 4.8, "caption": "Acid/carboxylate speciation model; approximate dilute-solution treatment."}}

# acid-reactions | Formation and acyl substitution

Activate the acid or use appropriate catalysis

## acid-reactions:rule | Core Rules

Primary alcohols and aldehydes can be oxidized to acids. Fischer esterification combines acid and alcohol under acid catalysis; removing water or using excess alcohol drives the reversible reaction. Direct acid plus amine usually first gives an ammonium carboxylate salt; efficient amide synthesis generally requires activation or dehydrating conditions.

## acid-reactions:cue | Memory Cues

Proton transfer is often faster than acyl substitution. Check it before drawing a new amide bond.

## acid-reactions:trap | Common Mistakes / MCAT Traps

OH− is a poor leaving group. Acid catalysis can enable water loss; coupling reagents or reactive acid derivatives can enable amide formation. Do not draw spontaneous amide synthesis just by mixing acid and amine in water.

## acid-reactions:app | MCAT Applications

Peptide synthesis uses activated carboxyl groups; enzyme-catalyzed biosynthesis likewise supplies activation and organization.

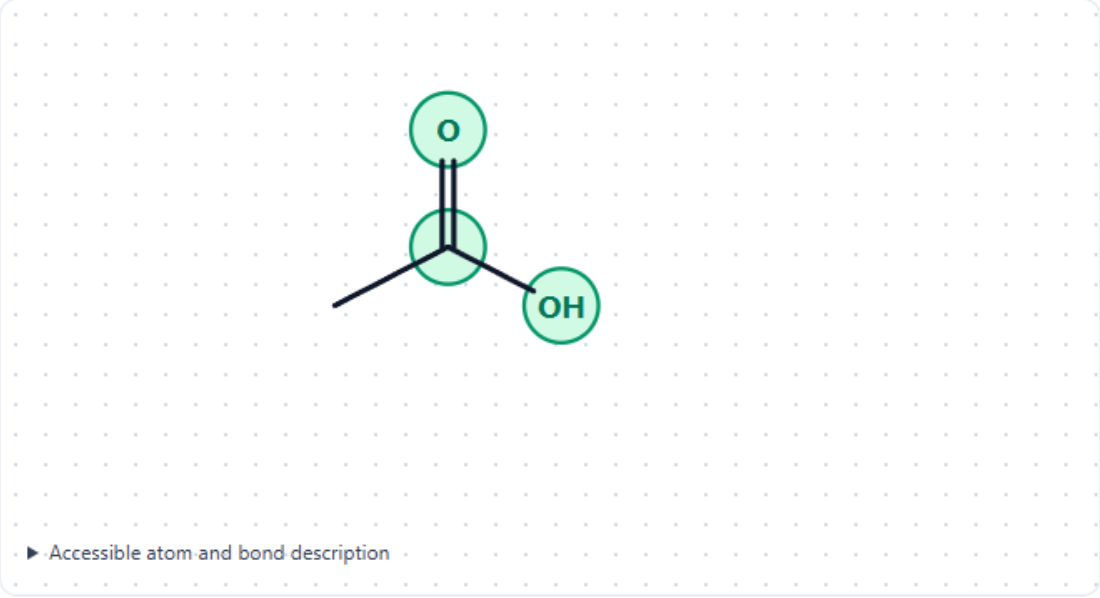
## acid-reactions:step-1 | 1. Acid substrate

The electrophilic acyl carbon is the carbonyl carbon. Protonating carbonyl oxygen increases electrophilicity in acid-catalyzed esterification.

Displayed molecule ID: aceticacid

{"highlights": {"prin": [1, 2, 3]}, "revealName": false}

Screenshot: acid-reactions-step-1



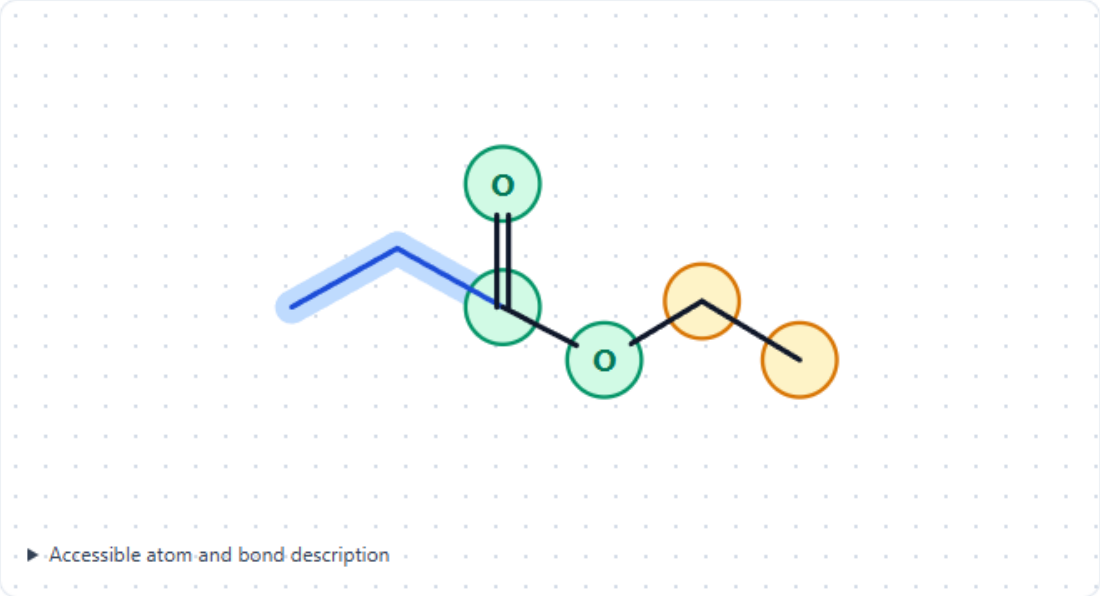

## acid-reactions:step-2 | 2. Ester product pattern

An ester has an OR group where the acid had OH. This example is ethyl propanoate: the O-ethyl and three-carbon acyl portions have different origins.

Displayed molecule ID: ester

{"highlights": {"path": [2, 1, 0], "prin": [2, 3, 4], "sub": [5, 6]}, "revealName": true}

Screenshot: acid-reactions-step-2



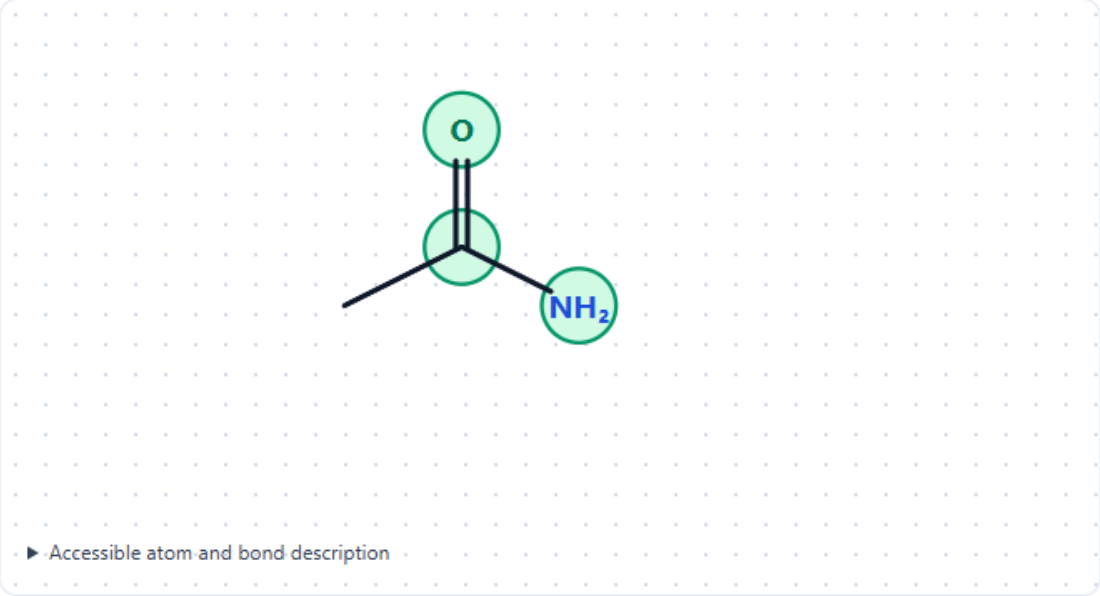

## acid-reactions:step-3 | 3. Amide product pattern

An amide has N attached directly to the acyl carbon. Formation requires conditions that overcome acid–base salt formation and poor leaving-group ability.

Displayed molecule ID: acetamide

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Screenshot: acid-reactions-step-3



# reduction-decarboxylation | Reduction and beta-keto-acid decarboxylation

Track carbon atoms, not just functional groups

## reduction-decarboxylation:rule | Core Rules

LiAlH4 followed by appropriate aqueous workup reduces a carboxylic acid to a primary alcohol. Ordinary NaBH4 conditions generally do not. Heating a beta-keto acid can release CO2 through a favorable cyclic transition state, initially giving an enol that tautomerizes to a carbonyl.

## reduction-decarboxylation:cue | Memory Cues

Reduction retains the acid carbon. Decarboxylation loses it as CO2.

## reduction-decarboxylation:trap | Common Mistakes / MCAT Traps

A beta-keto acid has a ketone at the beta position relative to the acid carbon. Not every carboxylic acid readily decarboxylates on heating. The six-membered arrangement is a transition state, not a stable ring intermediate.

## reduction-decarboxylation:app | MCAT Applications

Decarboxylation changes carbon count, a useful check when following metabolic or synthetic pathways.

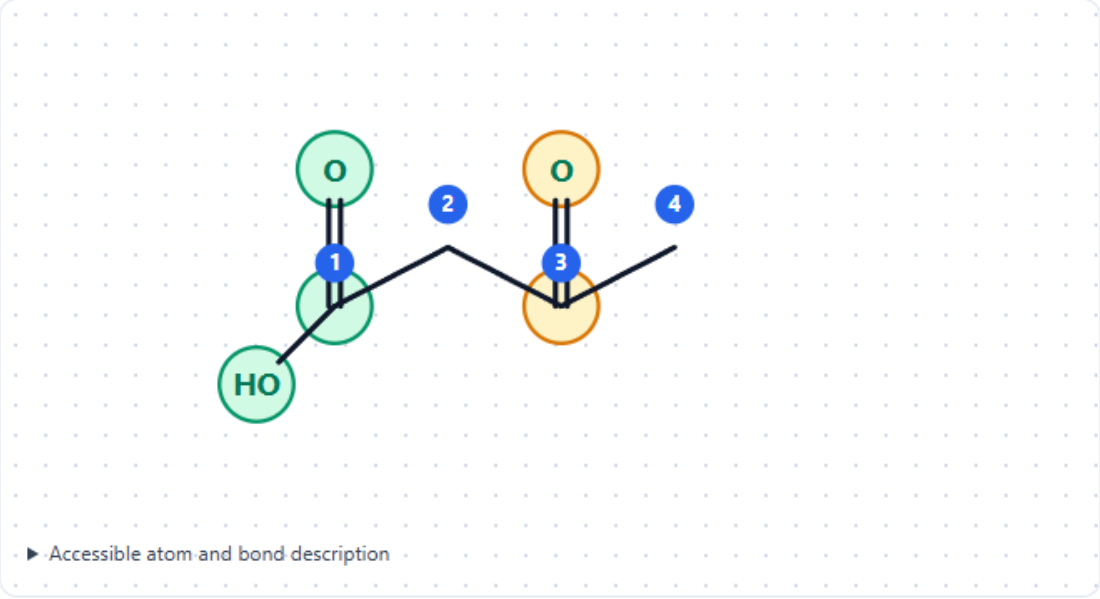
## reduction-decarboxylation:step-1 | 1. Number the beta-keto acid

C1 is the carboxyl carbon, C2 is alpha, and C3 is the ketone-bearing beta carbon.

Displayed molecule ID: acetoaceticacid

{"highlights": {"num": [0, 1, 2, 3], "prin": [0, 4, 5], "sub": [2, 6]}, "revealName": true}

Screenshot: reduction-decarboxylation-step-1



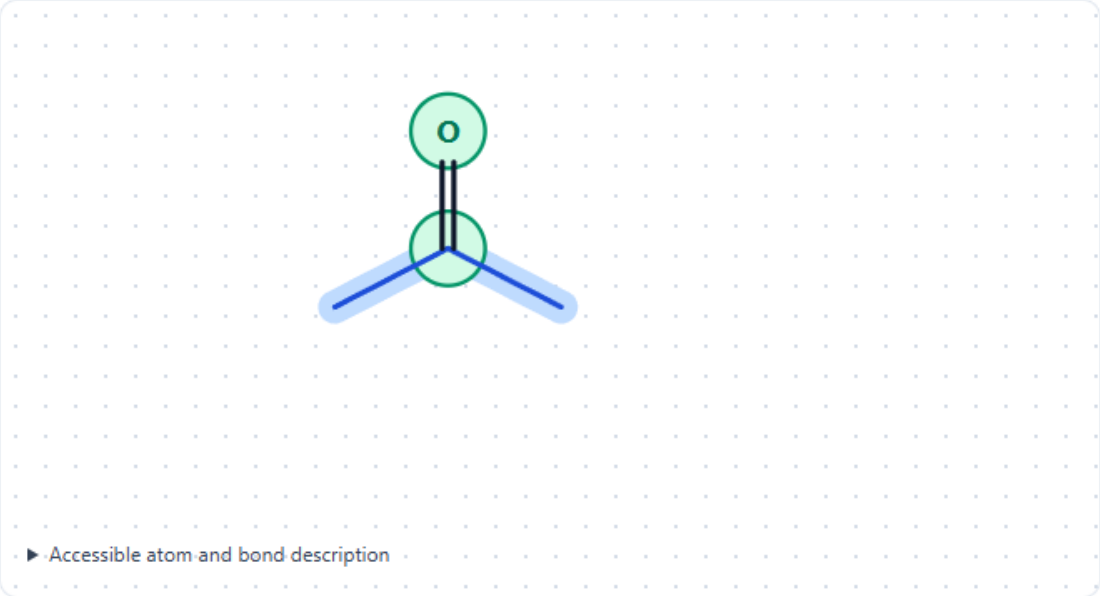

## reduction-decarboxylation:step-2 | 2. Lose the acid carbon

The carboxyl carbon leaves as CO2. The remaining three-carbon fragment initially forms an enol, then becomes acetone by tautomerization.

Displayed molecule ID: acetone

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Screenshot: reduction-decarboxylation-step-2



# soap | Saponification and amphiphiles

Hydrolysis produces carboxylate salts

## soap:rule | Core Rules

Saponification is base-promoted ester hydrolysis, including hydrolysis of triacylglycerols to glycerol and fatty-acid salts. A fatty-acid carboxylate has a polar ionic head and a nonpolar hydrocarbon tail, allowing micelle formation under suitable conditions.

## soap:cue | Memory Cues

Head interacts with water; tails gather away from water.

## soap:trap | Common Mistakes / MCAT Traps

Neutralizing a free fatty acid makes its salt, but that acid–base reaction is not ester hydrolysis. A micelle has hydrophobic material inside; it is not the same structure as a lipid bilayer.

## soap:app | MCAT Applications

Soap can disperse nonpolar material into water through aggregates. Aggregation depends on concentration, ionic conditions, and tail structure.

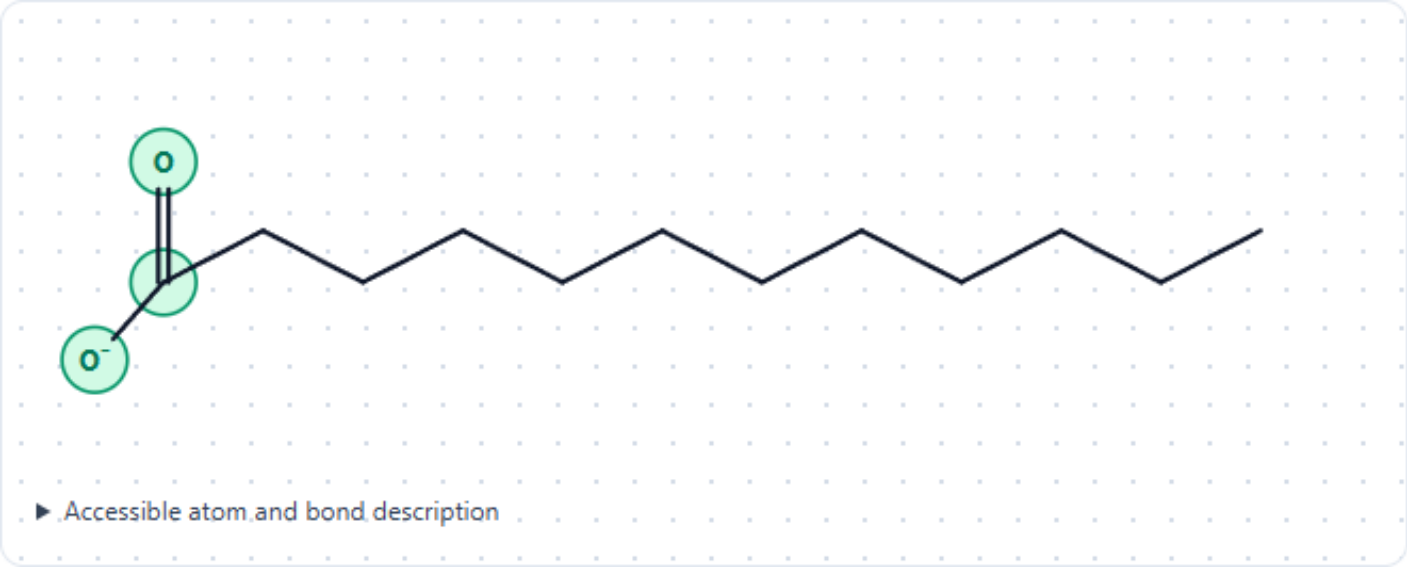
## soap:step-1 | 1. Find the head group

Laurate is a long-chain carboxylate. Its negative charge is represented on one O but delocalized over both. Counterions are omitted from this molecular-ion graph.

Displayed molecule ID: laurate

{"highlights": {"prin": [0, 12, 13]}, "revealName": true}

Screenshot: soap-step-1



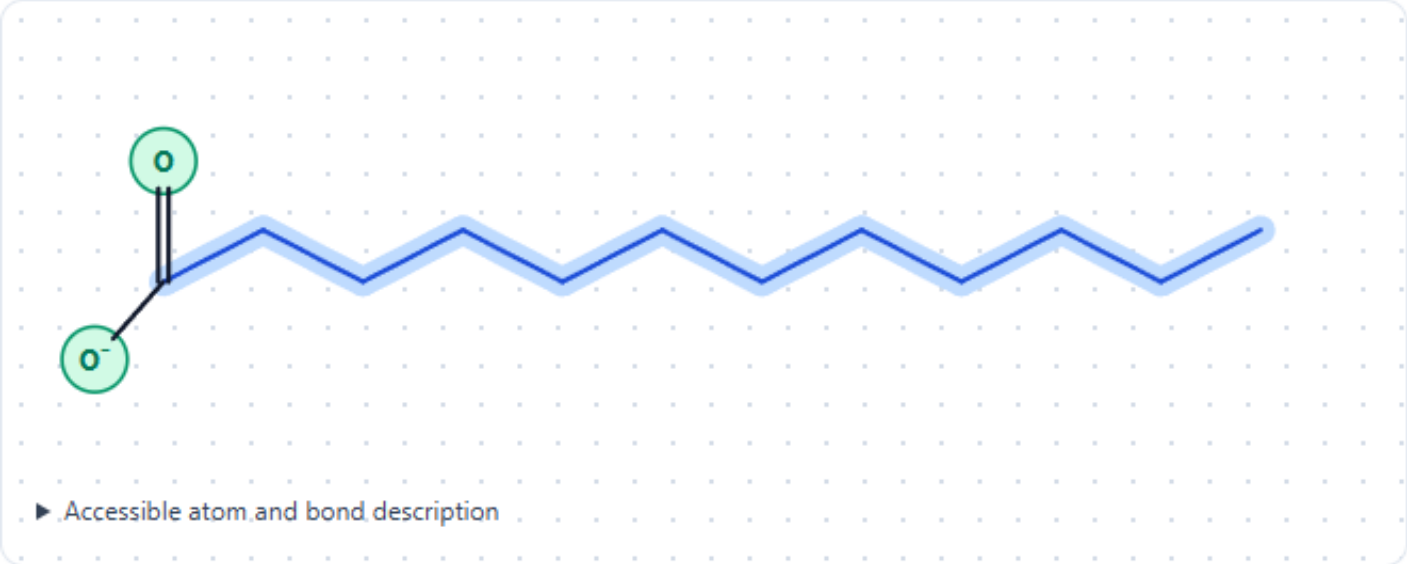

## soap:step-2 | 2. Contrast the tail

The hydrocarbon tail interacts favorably with other nonpolar material. Many such tails can cluster while ionic heads contact the surrounding water.

Displayed molecule ID: laurate

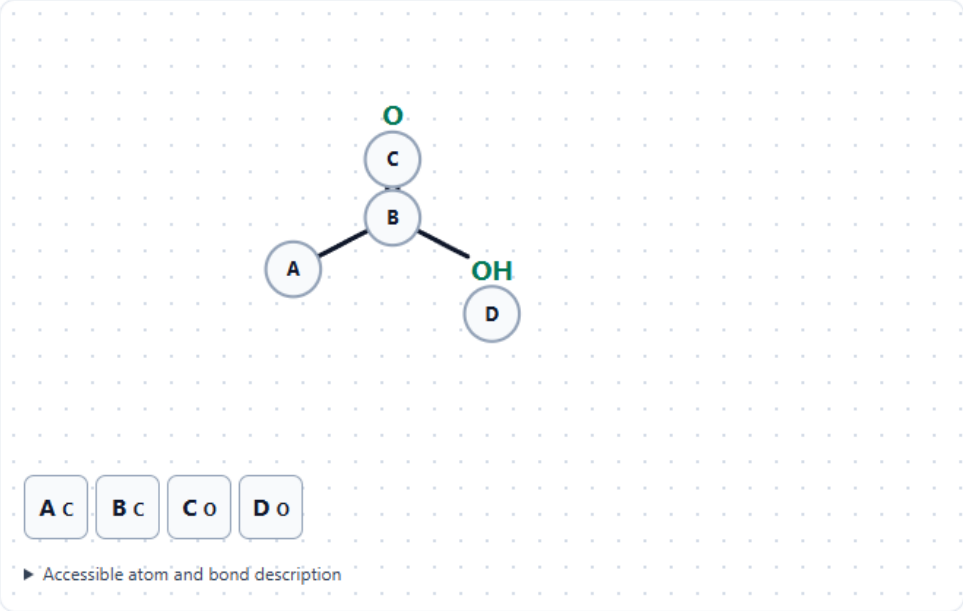
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Screenshot: soap-step-2



# acid-ratio | Calculate acid ionization

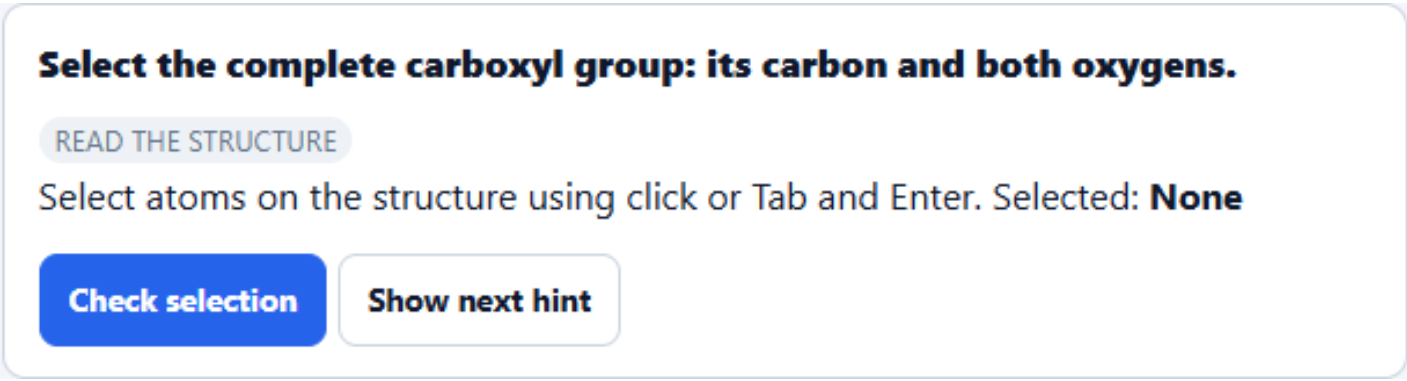
Screenshot: acid-ratio-stage-1-molecule




## acid-ratio:stage-1 | Select the complete carboxyl group: its carbon and both oxygens.

Answer type: atoms

Screenshot: acid-ratio-stage-1-question



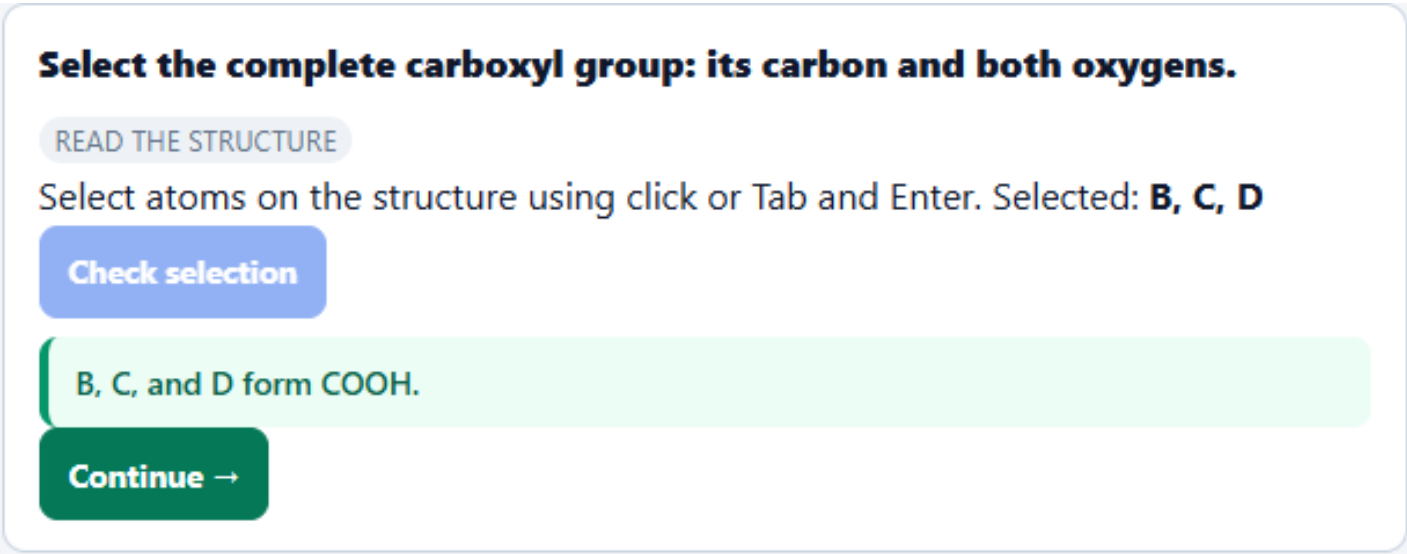

{"authorAnswer": [1, 2, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2, 3]}}

Hint 1: Include the shared carbon.

Hint 2: The methyl carbon is outside the group.

Explanation: B, C, and D form COOH.

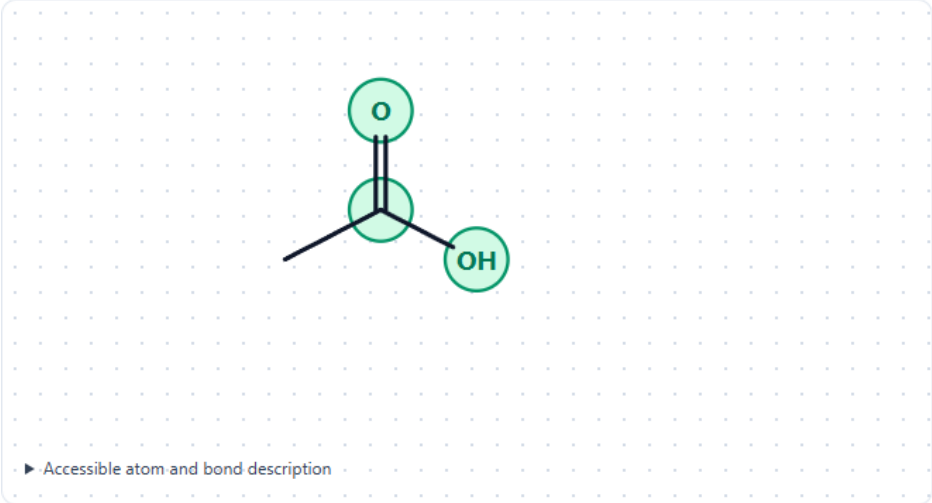
Screenshot: acid-ratio-stage-1-feedback



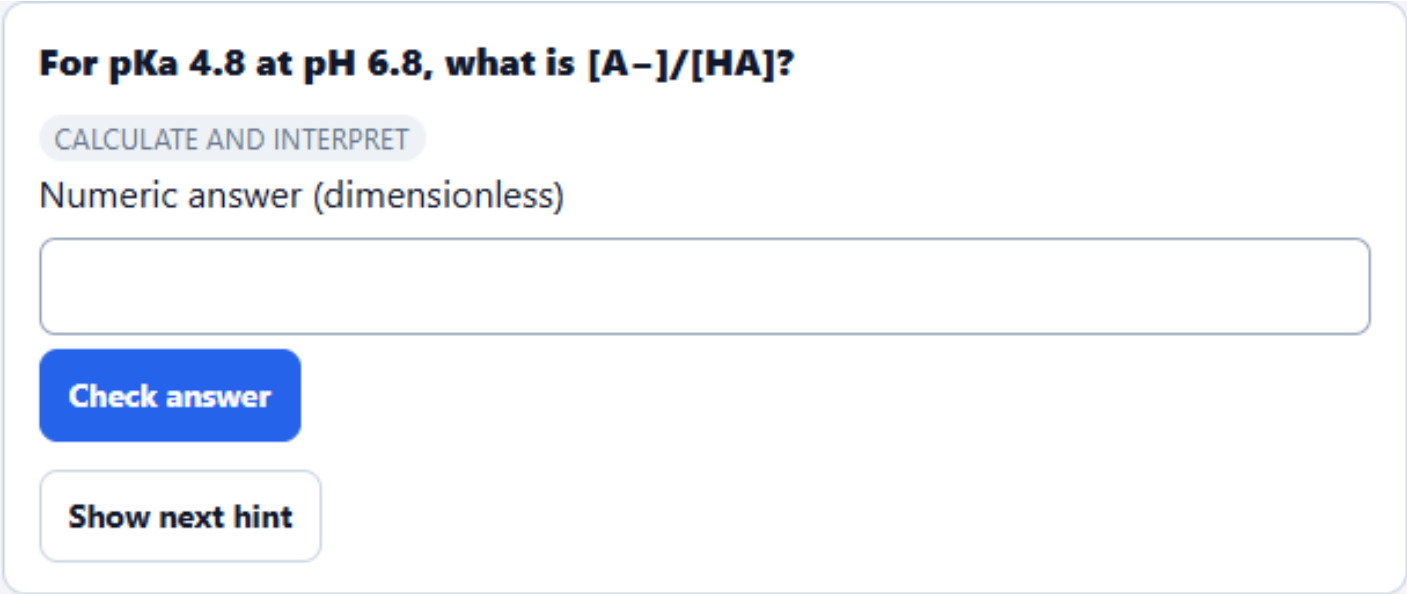

### acid-ratio:stage-2 | For pKa 4.8 at pH 6.8, what is [A−]/[HA]?

Answer type: number | unit: dimensionless

Screenshot: acid-ratio-stage-2-molecule



Screenshot: acid-ratio-stage-2-question



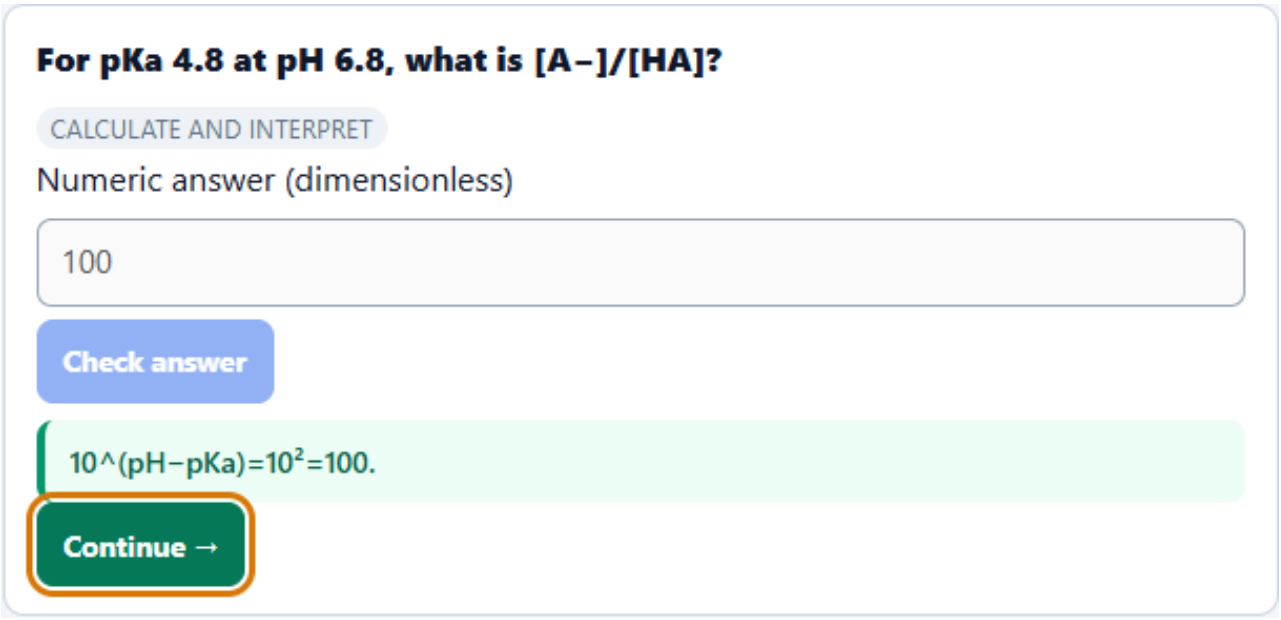

{"authorAnswer": 100, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use Henderson–Hasselbalch.

Hint 2: The pH exceeds pKa by two.

Explanation: 10^(pH−pKa)=10²=100.

Screenshot: acid-ratio-stage-2-feedback



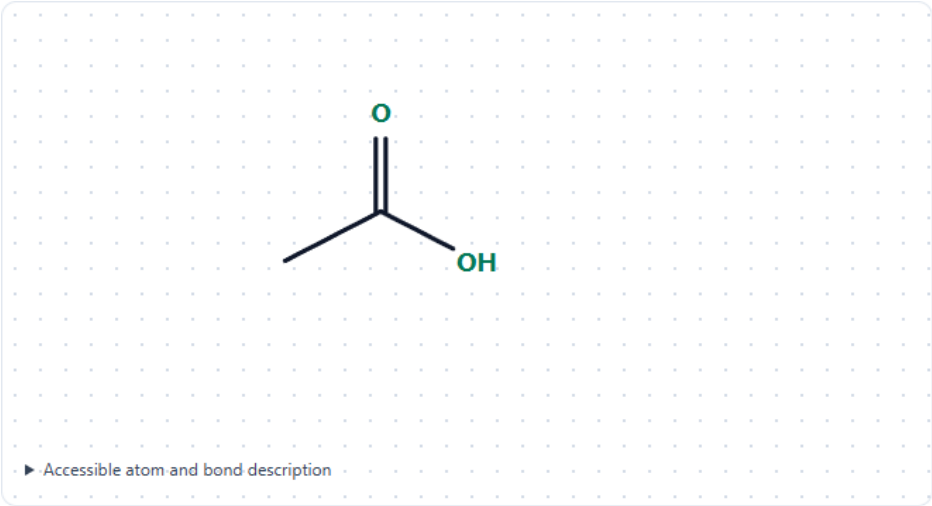

## acid-ratio:stage-3 | Which form predominates?

Answer type: choice

Option 0 (A): Neutral acid

Option 1 (B): Carboxylate

Screenshot: acid-ratio-stage-3-molecule




Screenshot: acid-ratio-stage-3-question



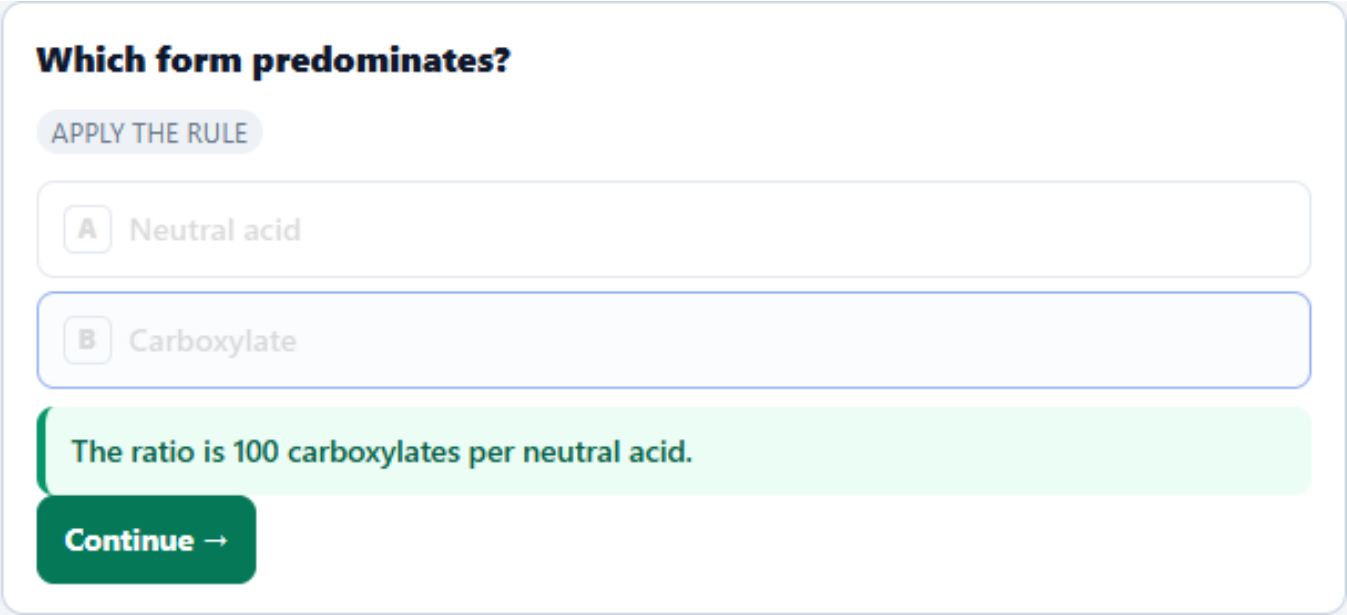

{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the calculated ratio.

Hint 2: Higher pH favors deprotonation.

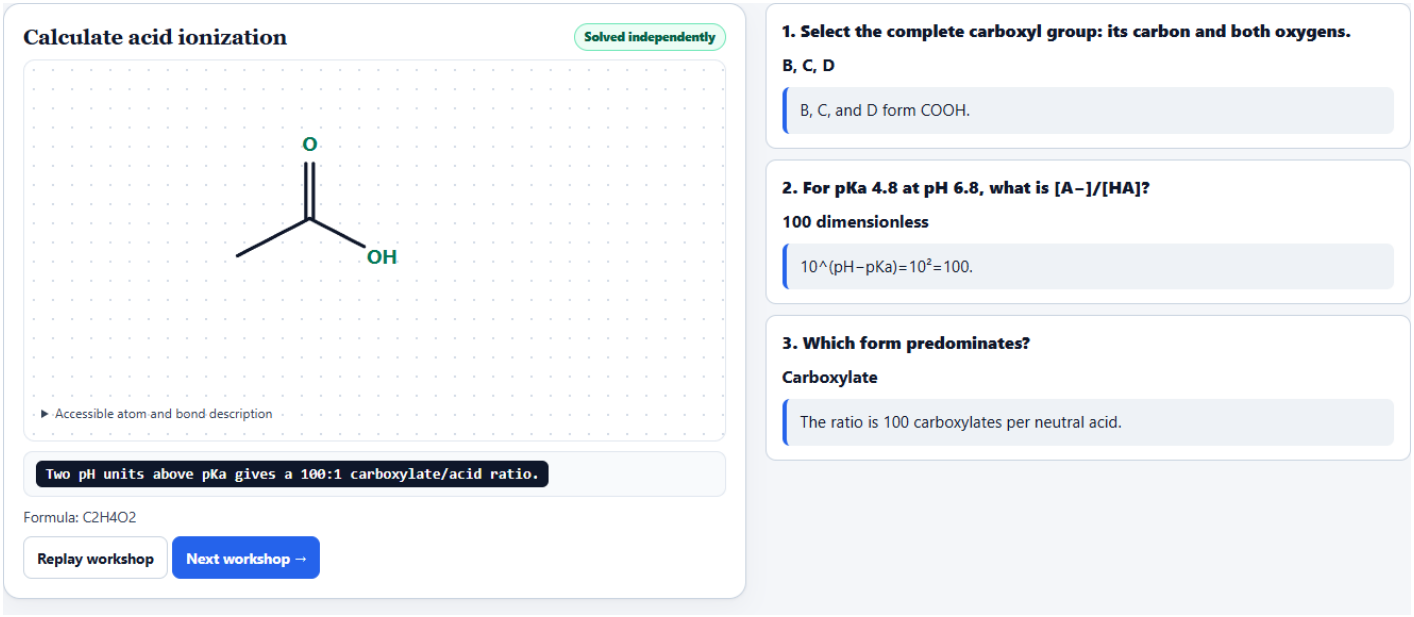
Explanation: The ratio is 100 carboxylates per neutral acid.

Screenshot: acid-ratio-stage-3-feedback



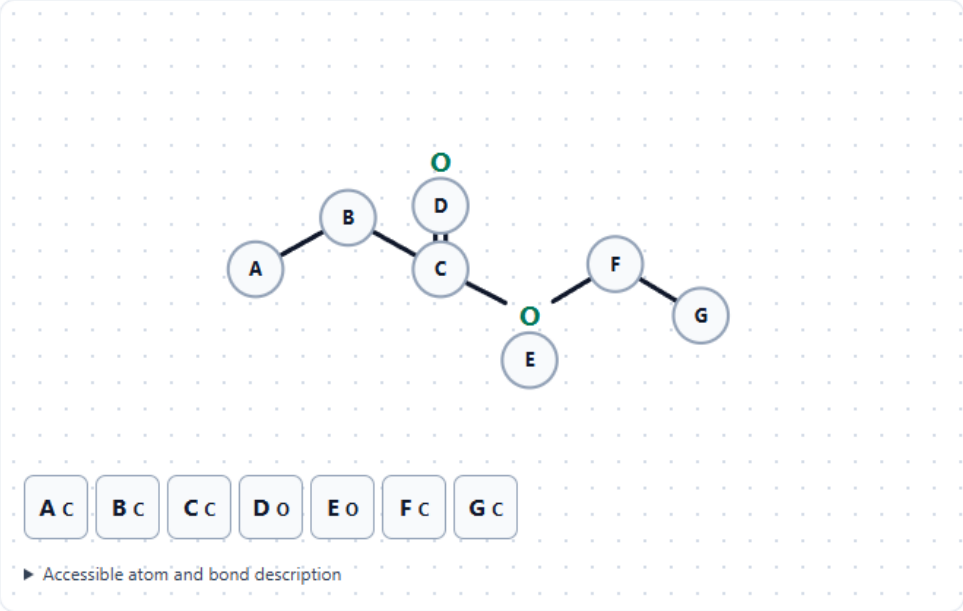

Completion summary: Two pH units above pKa gives a 100:1 carboxylate/acid ratio.

Screenshot: acid-ratio-complete



# make-ester | Reason about acid derivatives

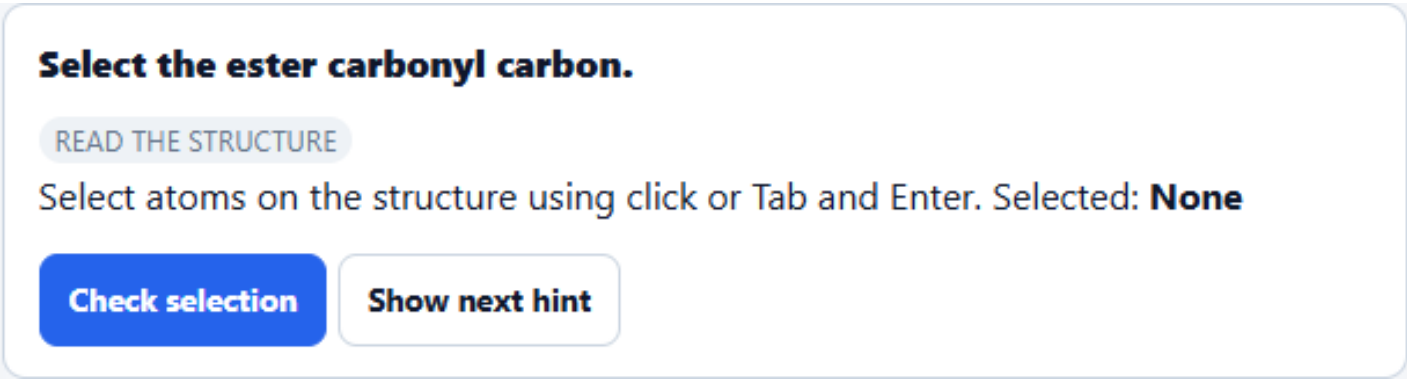
Screenshot: make-ester-stage-1-molecule




## make-ester:stage-1 | Select the ester carbonyl carbon.

Answer type: atoms

Screenshot: make-ester-stage-1-question



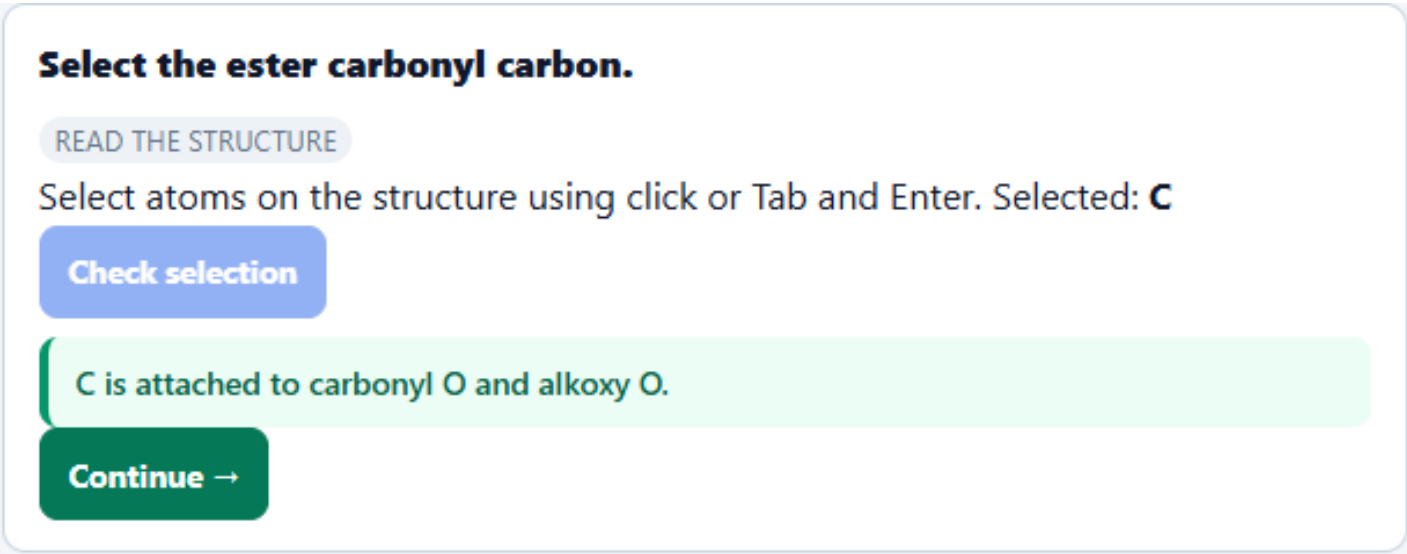

{"authorAnswer": [2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 3, 4]}}

Hint 1: Locate C(=O)–O–C.

Hint 2: Choose the shared carbon.

Explanation: C is attached to carbonyl O and alkoxy O.

Screenshot: make-ester-stage-1-feedback




### make-ester:stage-2 | Removing water from a Fischer esterification generally shifts toward…

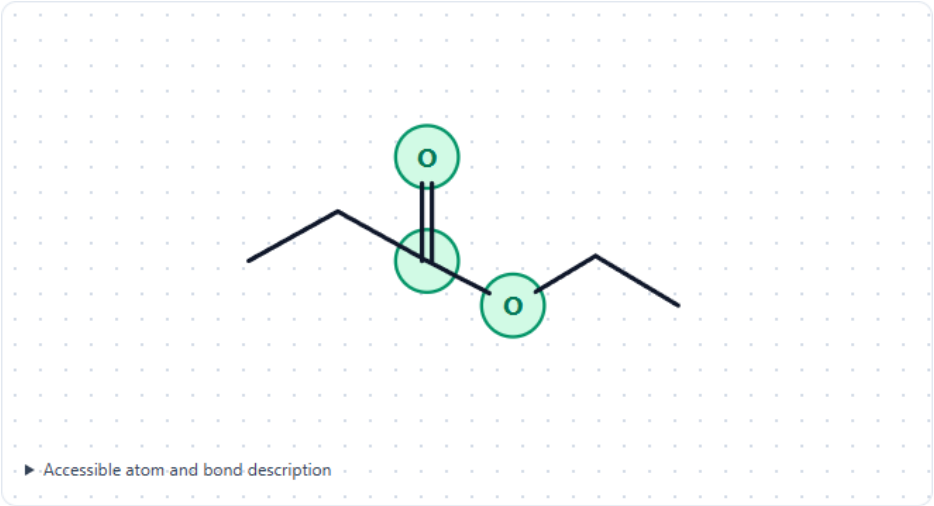
Answer type: choice

Option 0 (A): Ester

Option 1 (B): Acid and alcohol

Option 2 (C): No equilibrium change

Screenshot: make-ester-stage-2-molecule



Screenshot: make-ester-stage-2-question



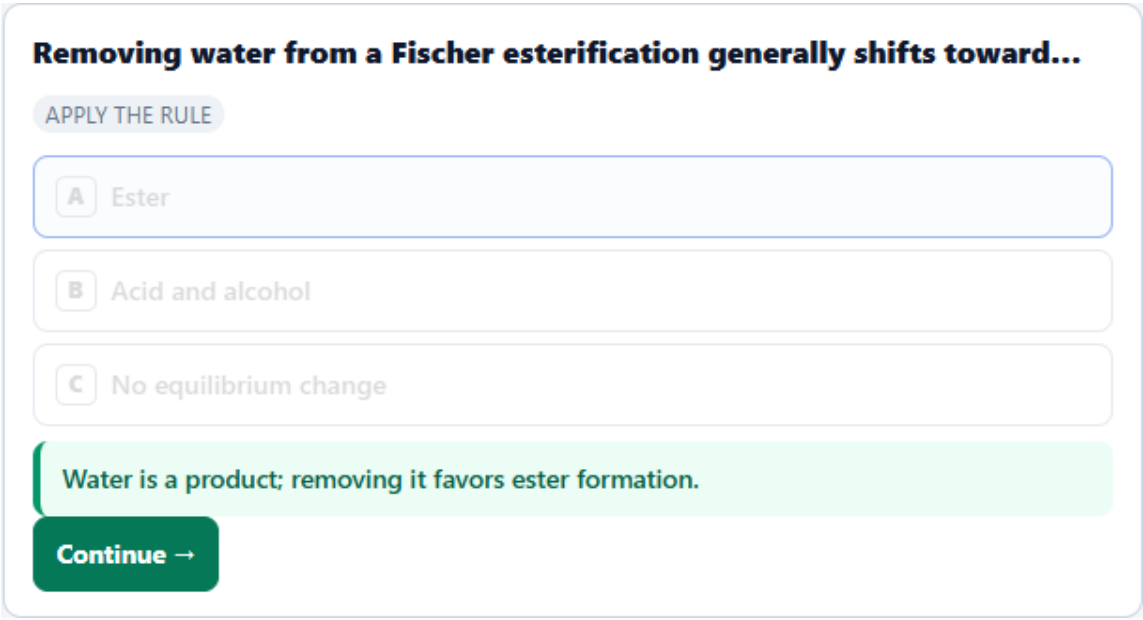

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Write acid + alcohol □ ester + water.

Hint 2: Apply Le Châtelier’s principle.

Explanation: Water is a product; removing it favors ester formation.

Screenshot: make-ester-stage-2-feedback




## make-ester:stage-3 | Mixing a carboxylic acid and an amine in water initially favors…

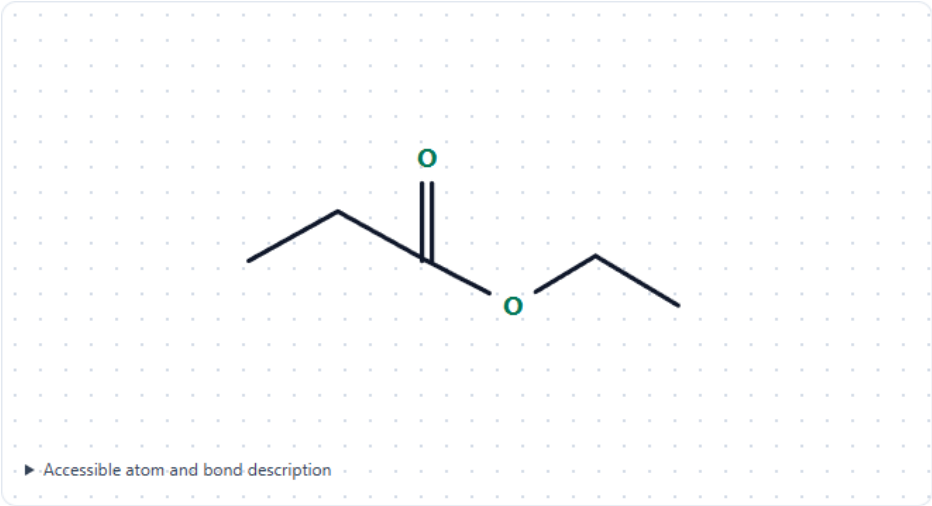
Answer type: choice

Option 0 (A): Ammonium carboxylate salt

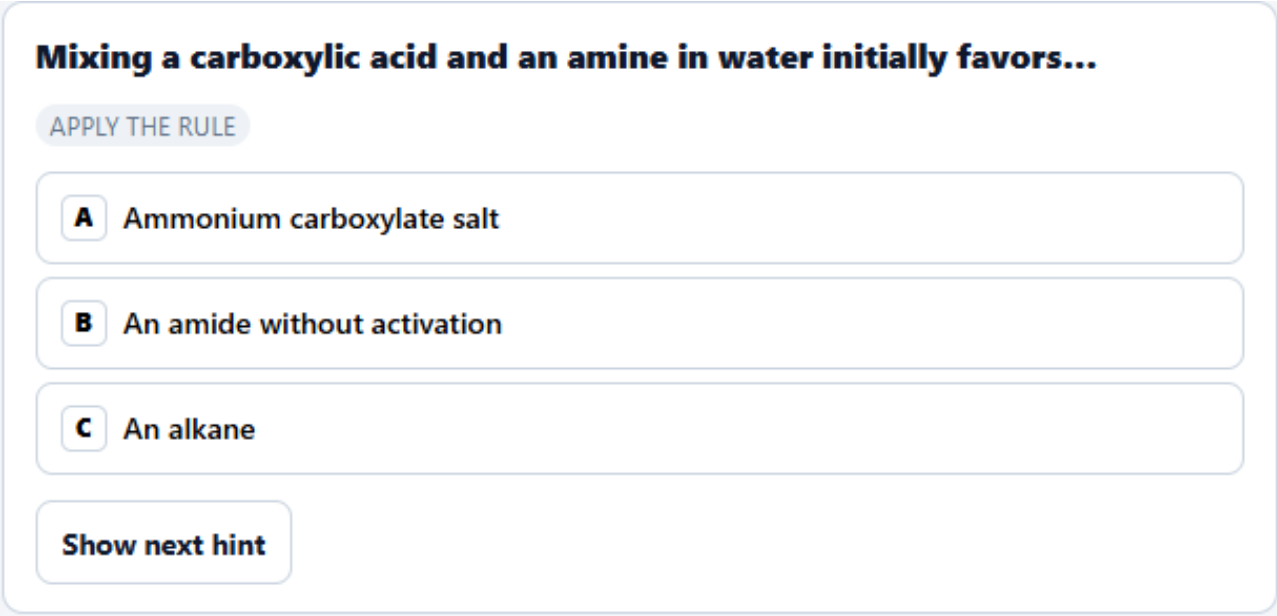
Option 1 (B): An amide without activation

Option 2 (C): An alkane

Screenshot: make-ester-stage-3-molecule




Screenshot: make-ester-stage-3-question



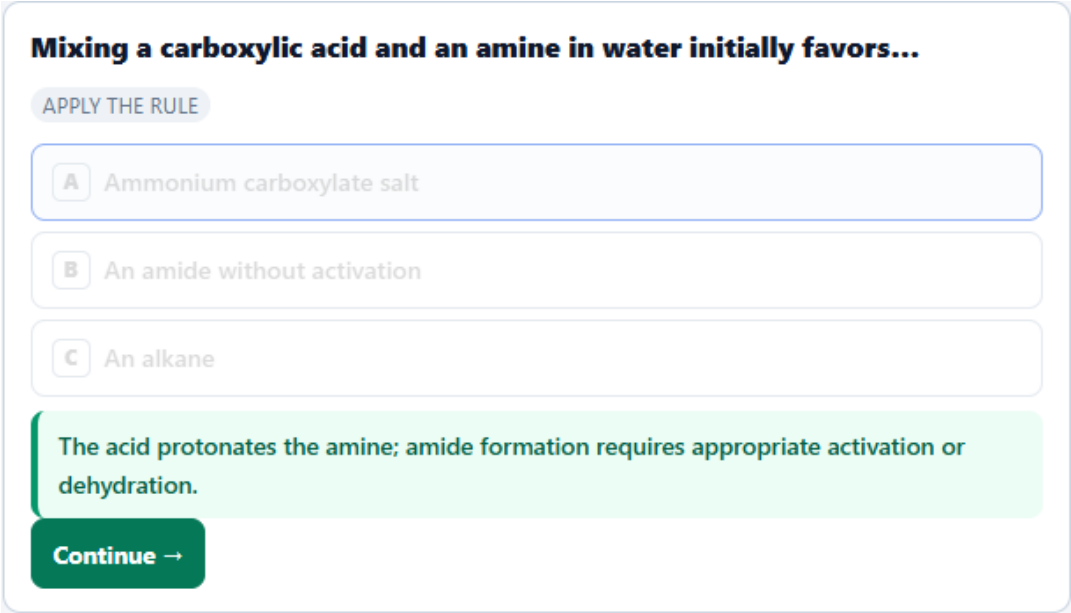

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider the fast acid–base step.

Hint 2: The amine is a base.

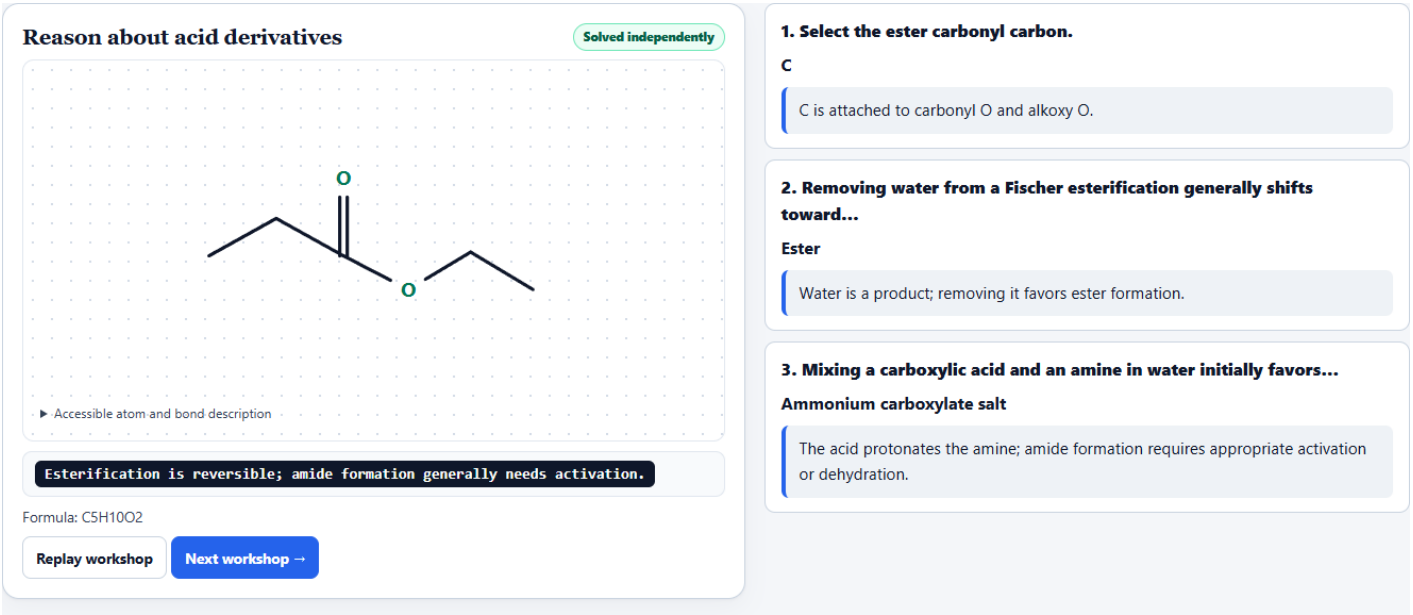
Explanation: The acid protonates the amine; amide formation requires appropriate activation or dehydration.

Screenshot: make-ester-stage-3-feedback



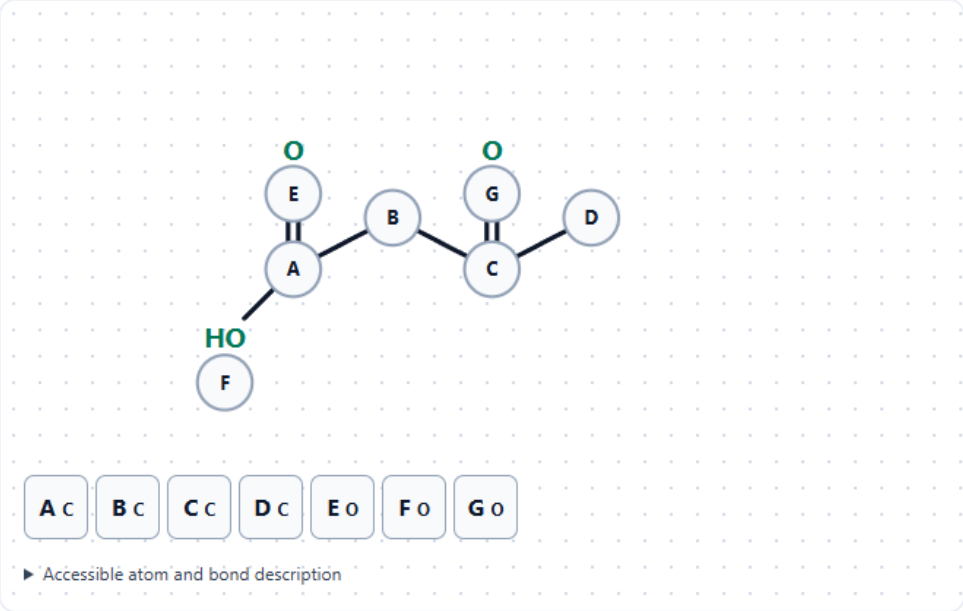

Completion summary: Esterification is reversible; amide formation generally needs activation.

Screenshot: make-ester-complete



# lose-co2 | Track decarboxylation carbons

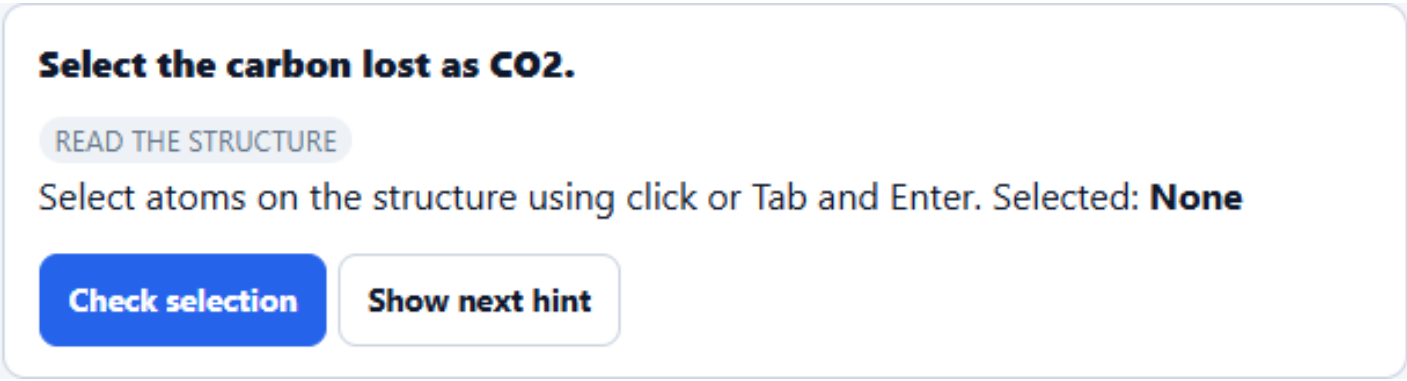
Screenshot: lose-co2-stage-1-molecule




## lose-co2:stage-1 | Select the carbon lost as CO2.

Answer type: atoms

Screenshot: lose-co2-stage-1-question



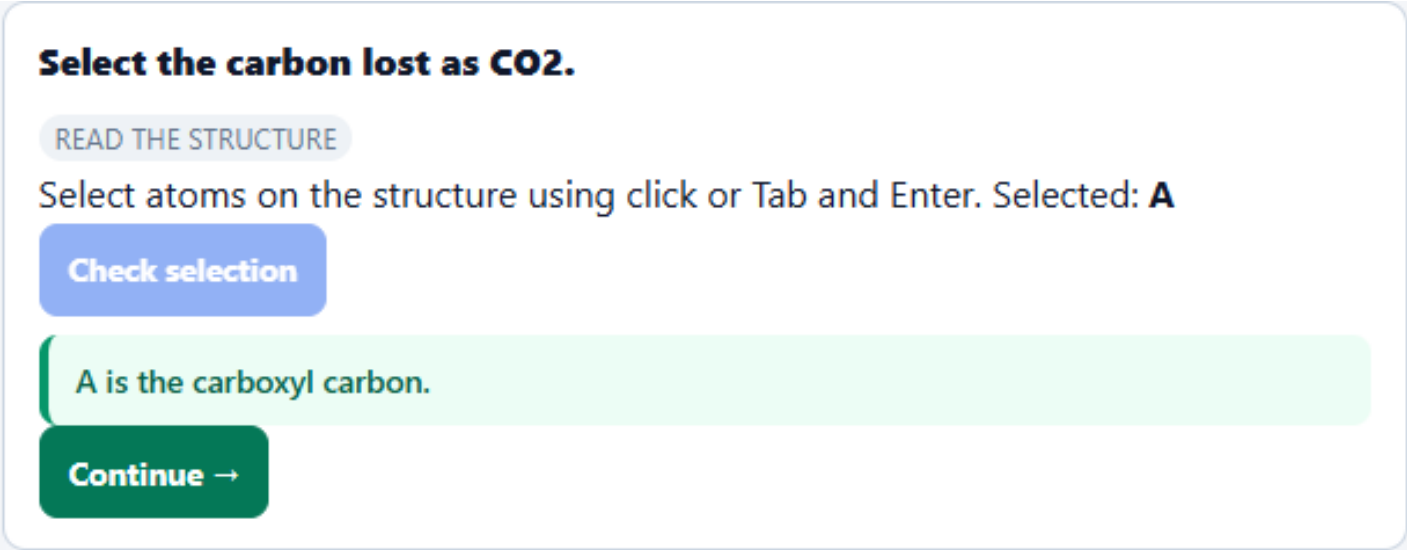

{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 4, 5]}}

Hint 1: The carboxyl group is expelled.

Hint 2: Do not choose the ketone carbon.

Explanation: A is the carboxyl carbon.

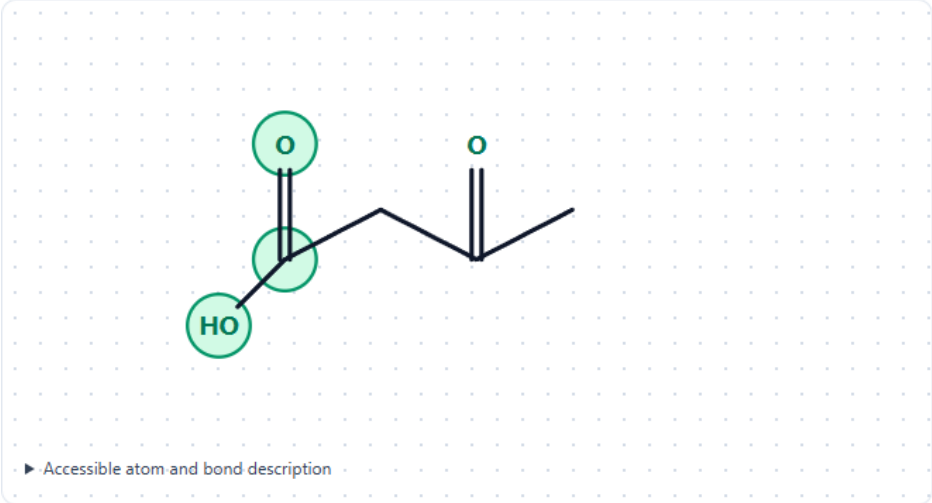
Screenshot: lose-co2-stage-1-feedback



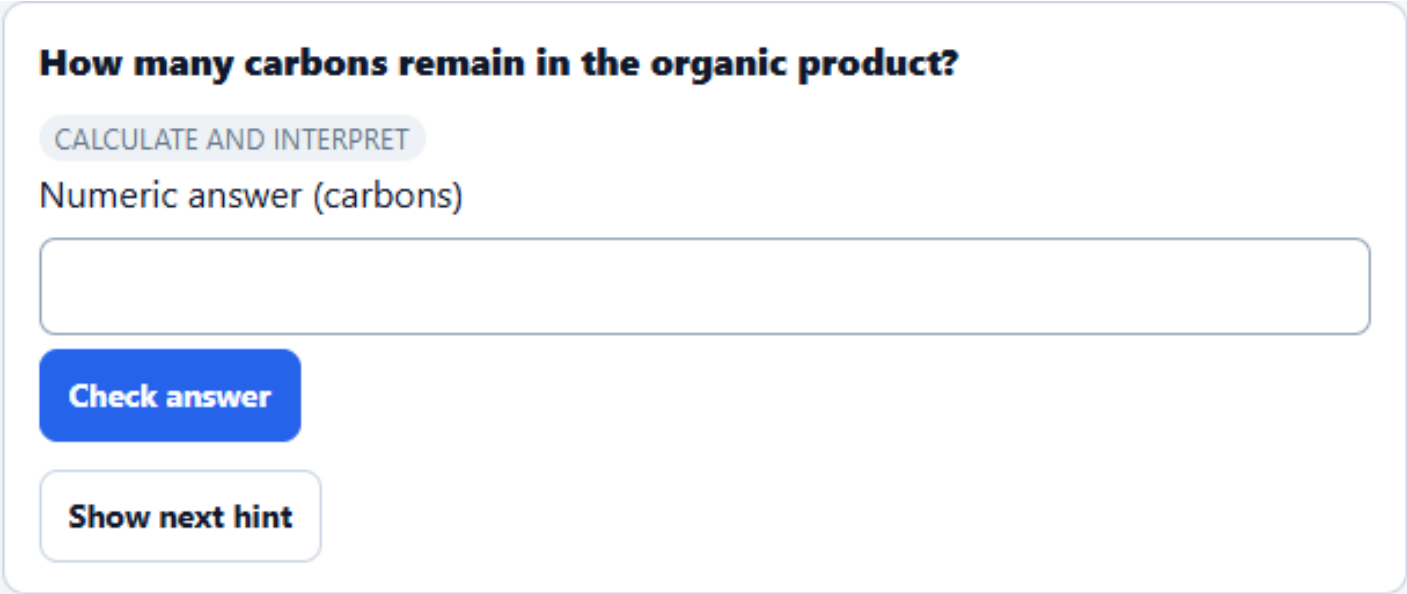

## lose-co2:stage-2 | How many carbons remain in the organic product?

Answer type: number | unit: carbons

Screenshot: lose-co2-stage-2-molecule



Screenshot: lose-co2-stage-2-question



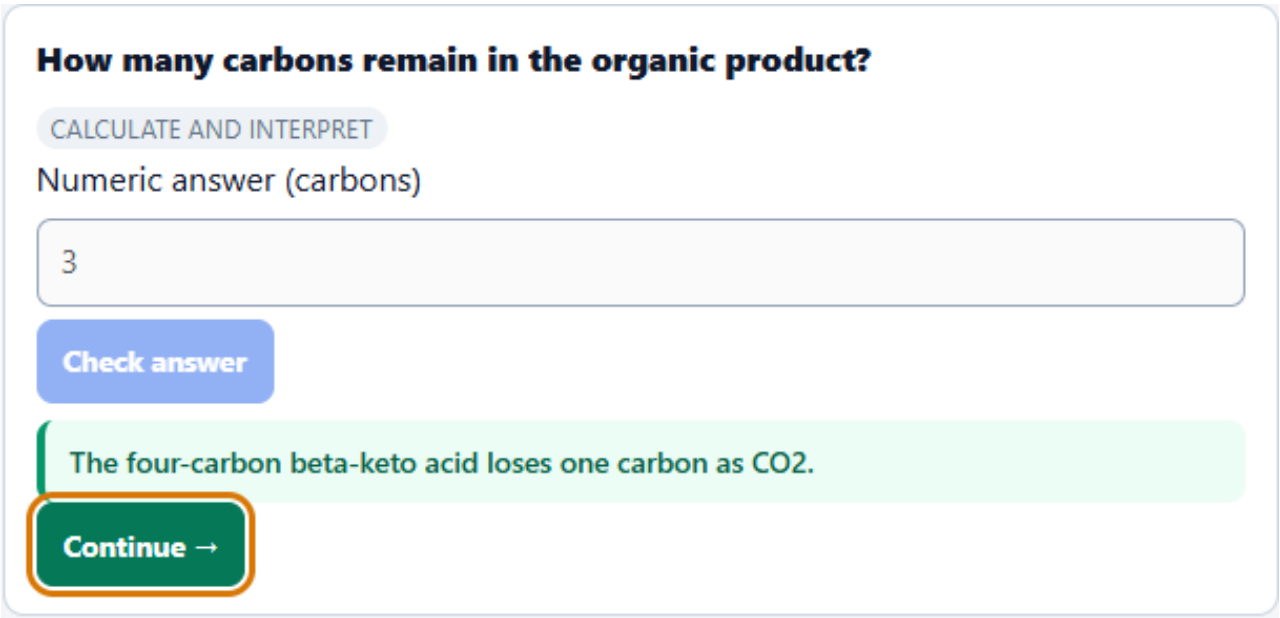

{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Subtract the departing carboxyl carbon.

Hint 2: 4 − 1 = ?

Explanation: The four-carbon beta-keto acid loses one carbon as CO2.

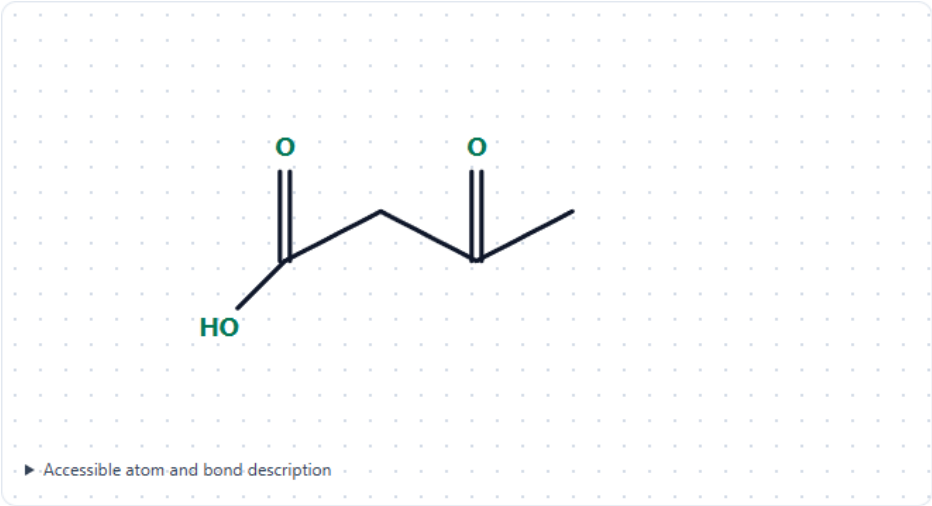
Screenshot: lose-co2-stage-2-feedback



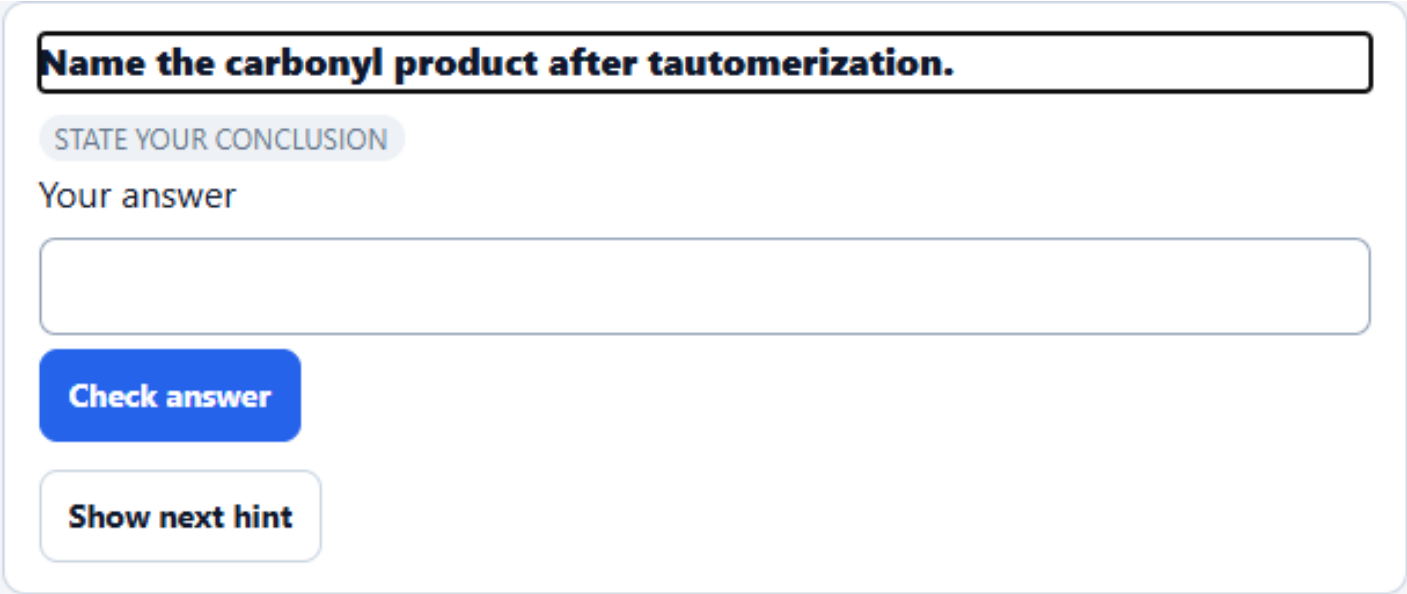

## lose-co2:stage-3 | Name the carbonyl product after tautomerization.

Answer type: text

Screenshot: lose-co2-stage-3-molecule




Screenshot: lose-co2-stage-3-question



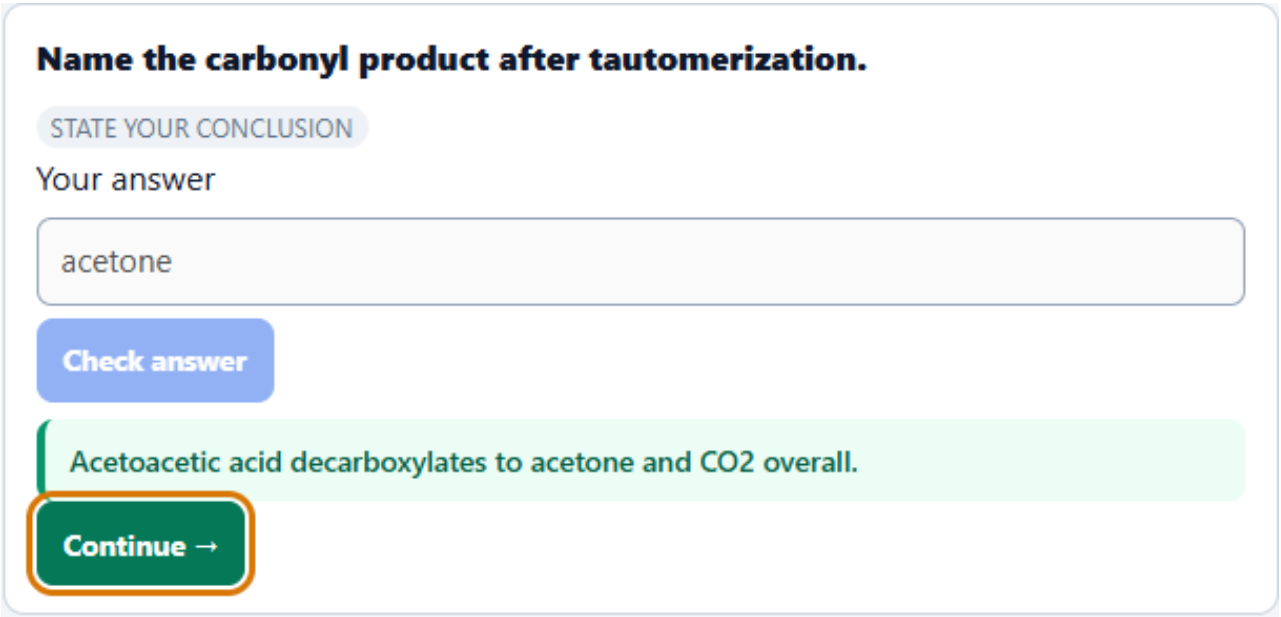

{"authorAnswer": ["acetone", "propanone", "propan-2-one"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: The remaining chain has three carbons.

Hint 2: The carbonyl becomes the middle carbon of that chain.

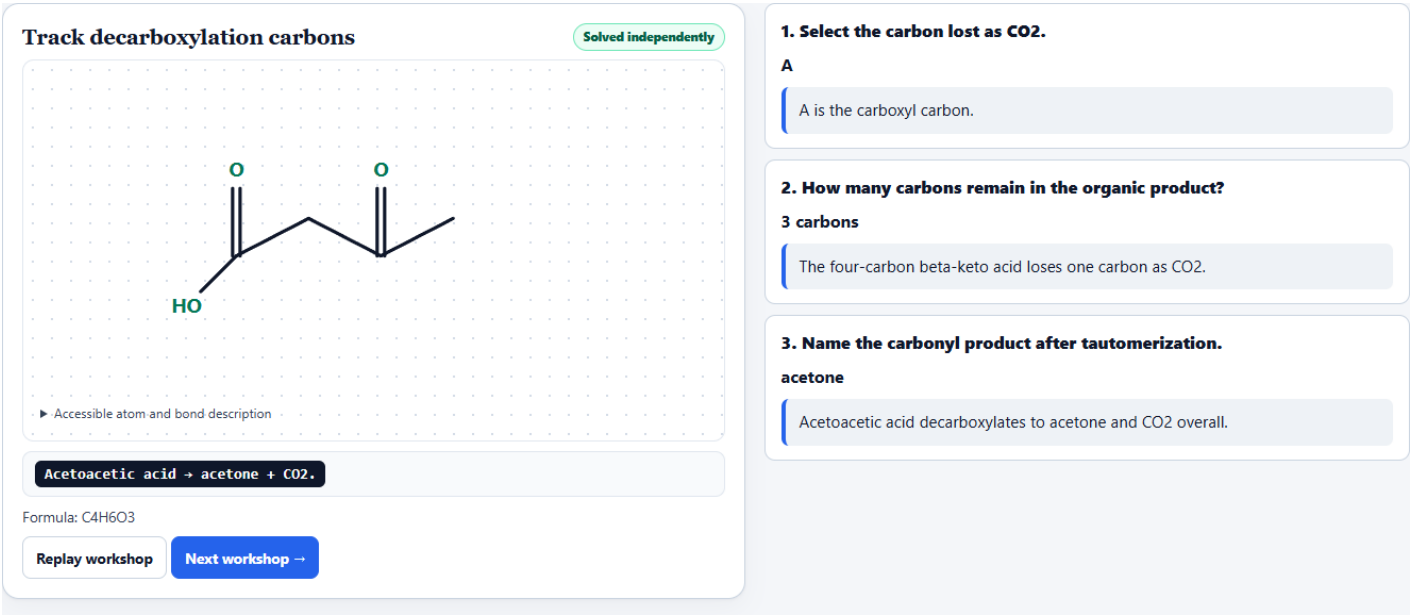
Explanation: Acetoacetic acid decarboxylates to acetone and CO2 overall.

Screenshot: lose-co2-stage-3-feedback



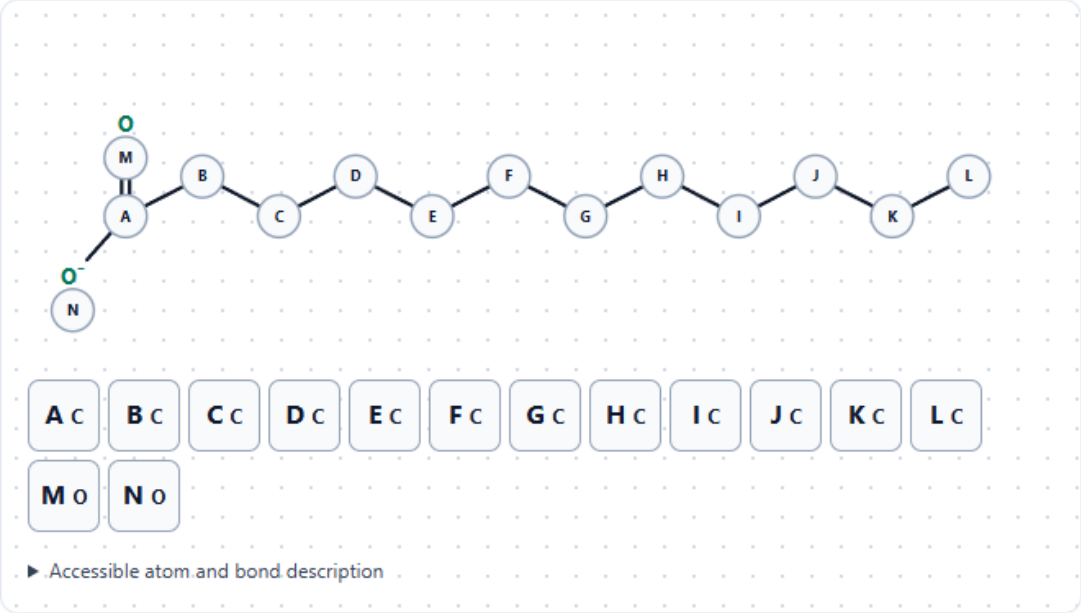

Completion summary: Acetoacetic acid → acetone + CO2.

Screenshot: lose-co2-complete



# soap-head | Connect structure and aggregation

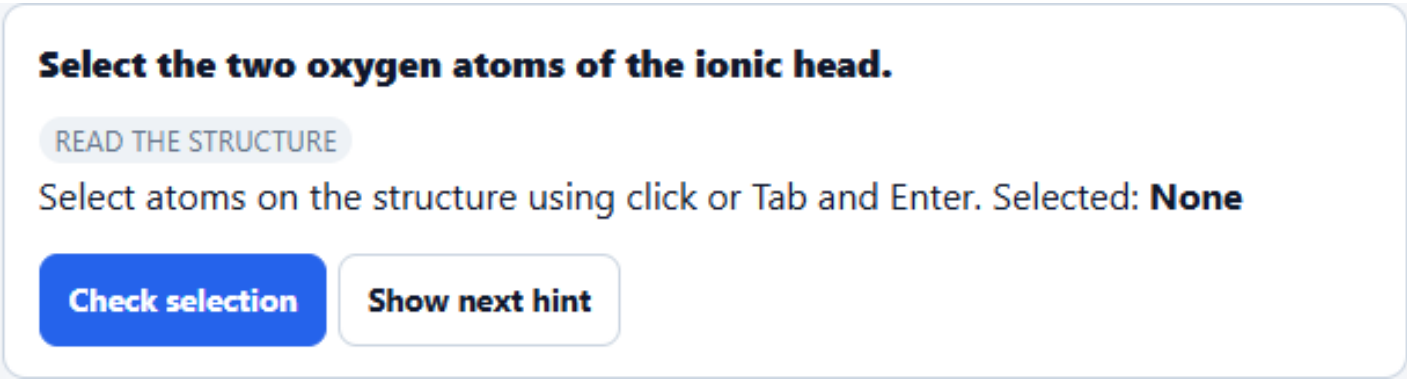
Screenshot: soap-head-stage-1-molecule




## soap-head:stage-1 | Select the two oxygen atoms of the ionic head.

Answer type: atoms

Screenshot: soap-head-stage-1-question



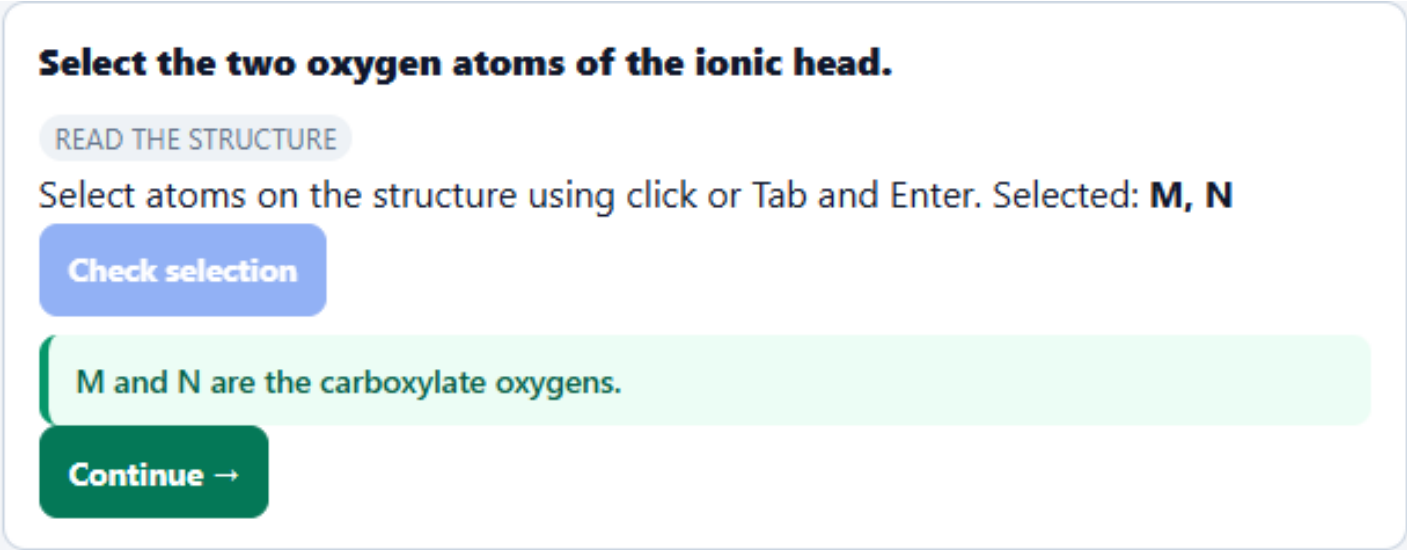

{"authorAnswer": [12, 13], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [12, 13]}}

Hint 1: Locate the O labels at the end of the chain.

Hint 2: The charge is delocalized across the two oxygens.

Explanation: M and N are the carboxylate oxygens.

Screenshot: soap-head-stage-1-feedback



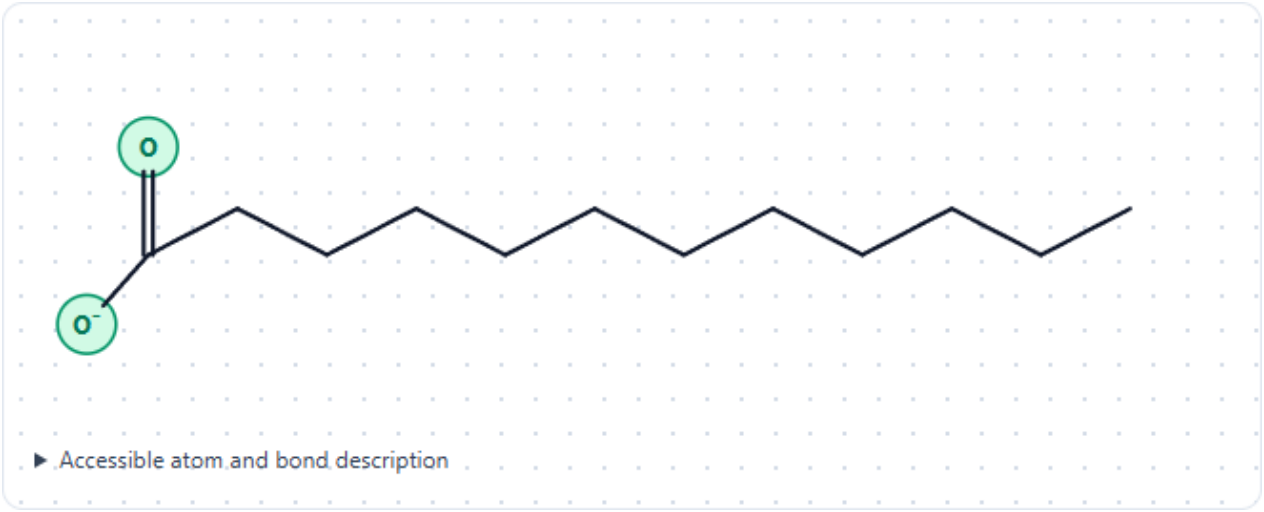

## soap-head:stage-2 | In an aqueous micelle, hydrocarbon tails tend to point…

Answer type: choice

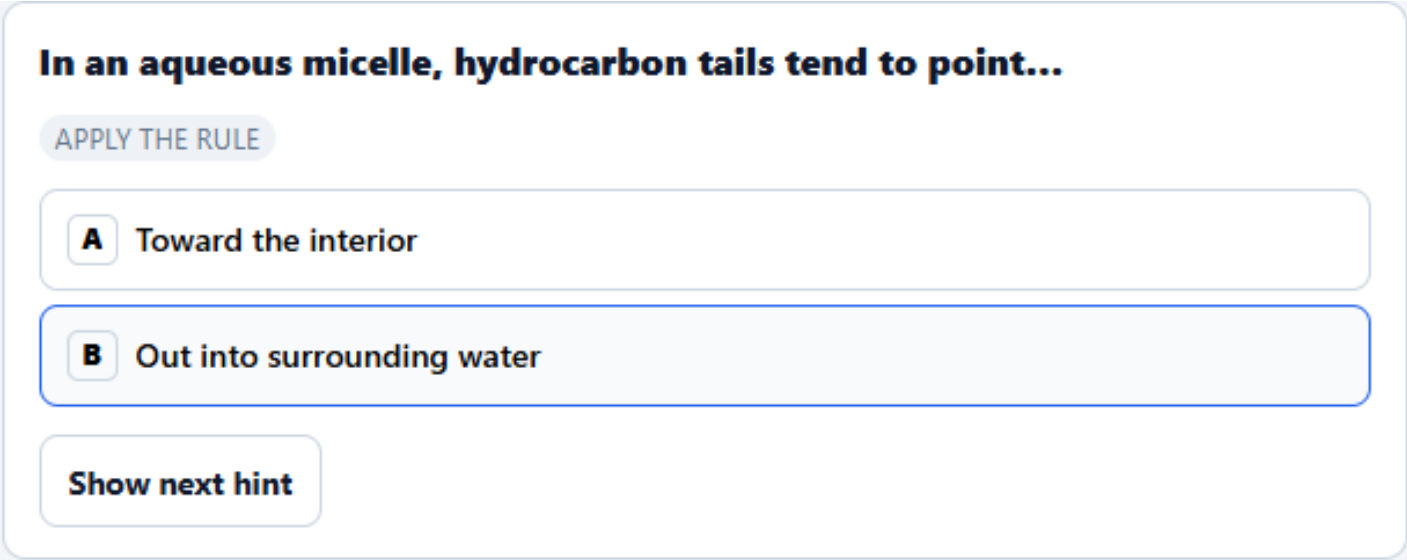
Option 0 (A): Toward the interior

Option 1 (B): Out into surrounding water

Screenshot: soap-head-stage-2-molecule



Screenshot: soap-head-stage-2-question



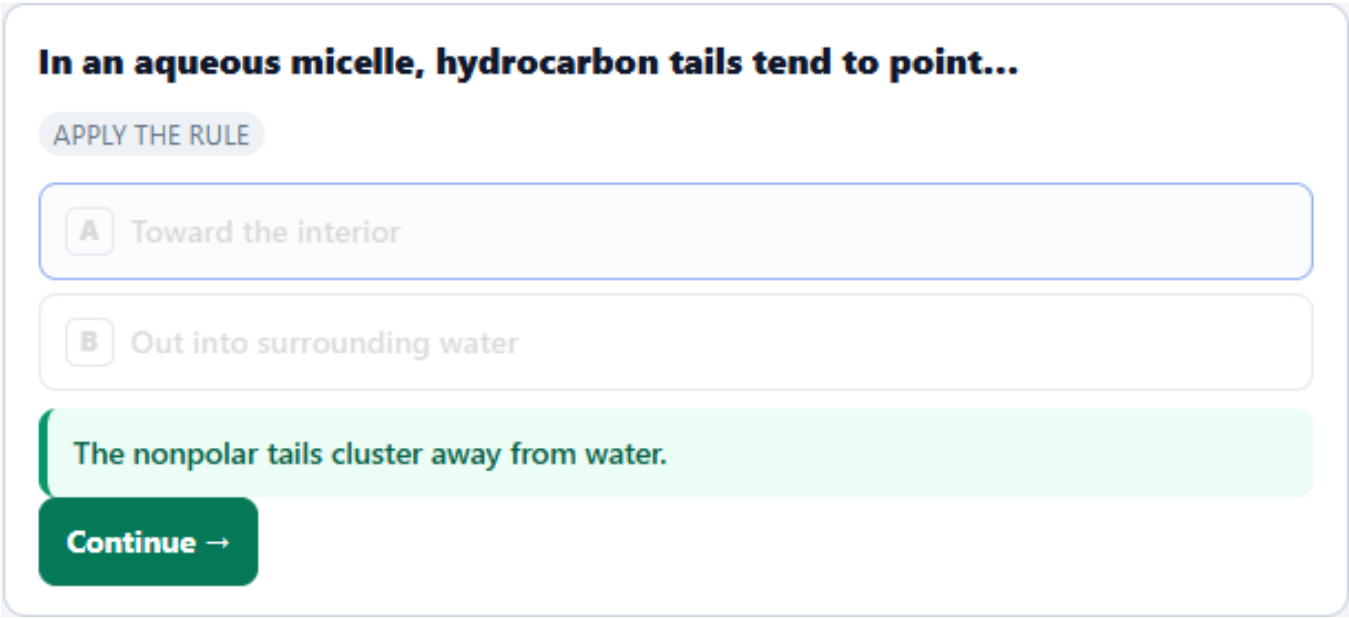

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish polar and nonpolar regions.

Hint 2: Ionic heads remain exposed to water.

Explanation: The nonpolar tails cluster away from water.

Screenshot: soap-head-stage-2-feedback




## soap-head:stage-3 | Saponification of a triacylglycerol produces…

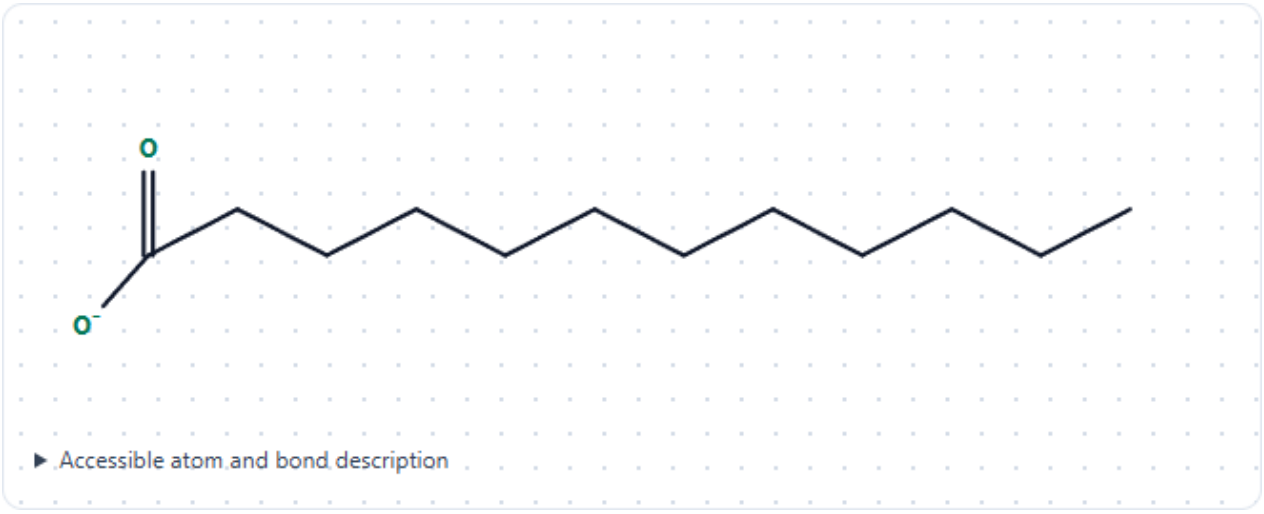
Answer type: choice

Option 0 (A): Glycerol and fatty-acid salts

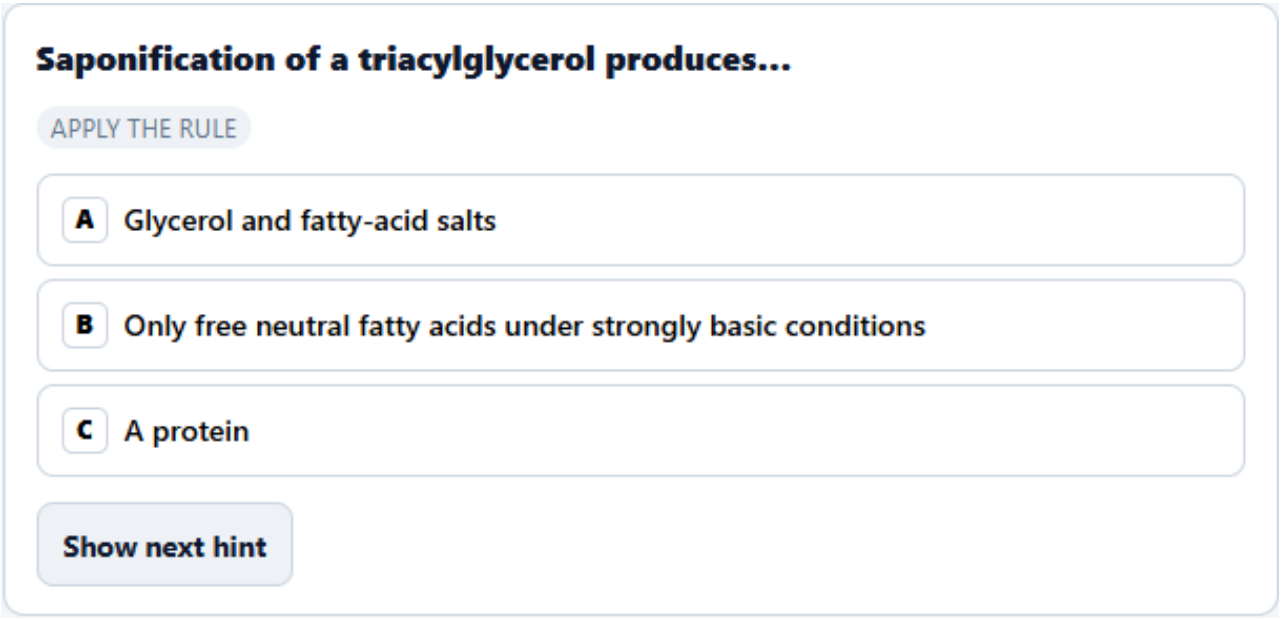
Option 1 (B): Only free neutral fatty acids under strongly basic conditions

Option 2 (C): A protein

Screenshot: soap-head-stage-3-molecule




Screenshot: soap-head-stage-3-question



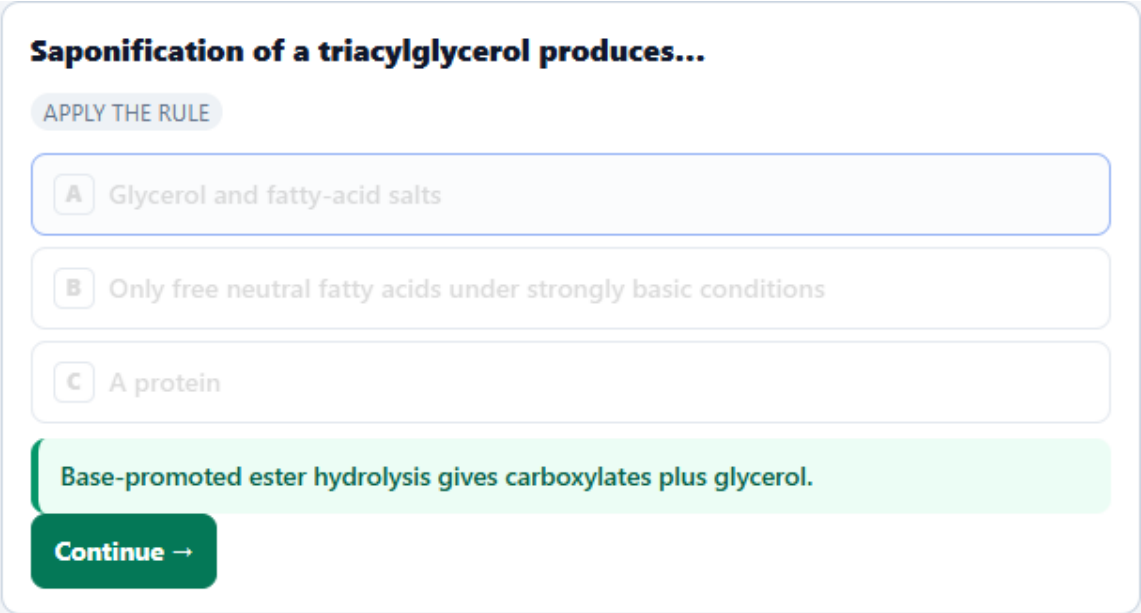

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hydrolyze all three ester links.

Hint 2: Under basic conditions the acids are deprotonated.

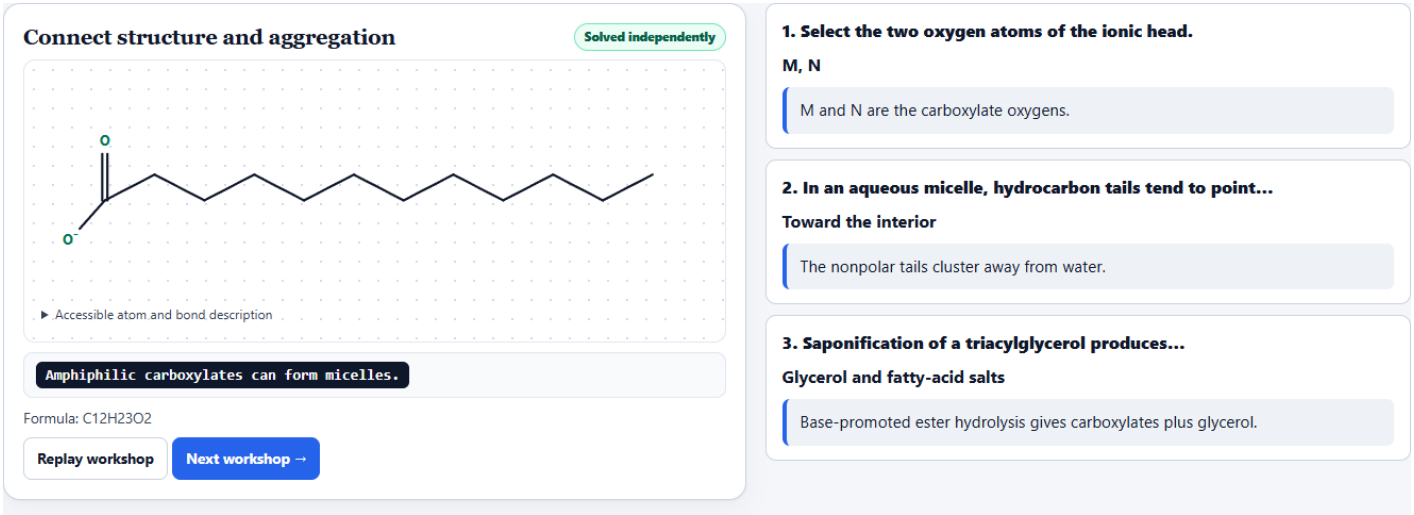
Explanation: Base-promoted ester hydrolysis gives carboxylates plus glycerol.

Screenshot: soap-head-stage-3-feedback




Completion summary: Amphiphilic carboxylates can form micelles.

Screenshot: soap-head-complete



# Molecular repository: author claims to verify

## aceticacid

Author claims: ethanoic acid | formula C2H4O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

## acetate

Author claims: acetate (one resonance contributor) | formula C2H3O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

## ester

Author claims: ethyl propanoate | formula C5H10O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=156.00, y=52.00; charge=0; explicit H override=none

4 (E): O; x=208.00, y=142.00; charge=0; explicit H override=none

5 (F): C; x=258.00, y=112.00; charge=0; explicit H override=none

6 (G): C; x=308.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1]]

## acetamide

Author claims: ethanamide | formula C2H5NO

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

## acetoaceticacid

Author claims: 3-oxobutanoic acid (acetoacetic acid) | formula C4H6O3

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=50.00, y=45.00; charge=0; explicit H override=none

5 (F): O; x=10.00, y=155.00; charge=0; explicit H override=none

6 (G): O; x=166.00, y=45.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [0, 4, 2], [0, 5, 1], [2, 6, 2]]

## acetone

Author claims: propanone | formula C3H6O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

## laurate

Author claims: dodecanoate (laurate) | formula C12H23O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=282.00, y=115.00; charge=0; explicit H override=none

5 (F): C; x=340.00, y=85.00; charge=0; explicit H override=none

6 (G): C; x=398.00, y=115.00; charge=0; explicit H override=none

7 (H): C; x=456.00, y=85.00; charge=0; explicit H override=none

8 (I): C; x=514.00, y=115.00; charge=0; explicit H override=none

9 (J): C; x=572.00, y=85.00; charge=0; explicit H override=none

10 (K): C; x=630.00, y=115.00; charge=0; explicit H override=none

11 (L): C; x=688.00, y=85.00; charge=0; explicit H override=none

12 (M): O; x=50.00, y=45.00; charge=0; explicit H override=none

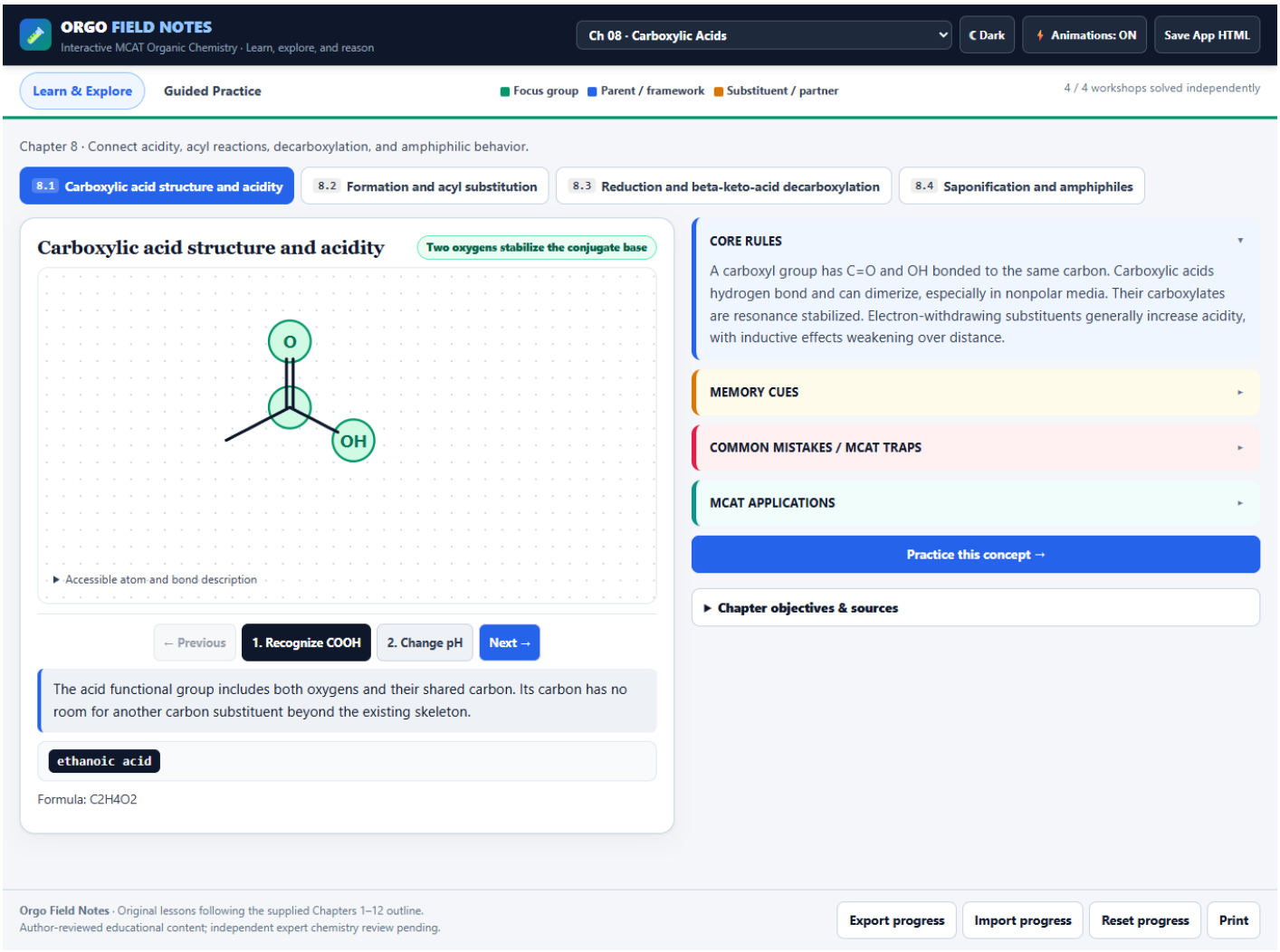
13 (N): O; x=10.00, y=160.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 1], [7, 8, 1], [8, 9, 1], [9, 10, 1], [10, 11, 1], [0, 12, 2], [0, 13, 1]]

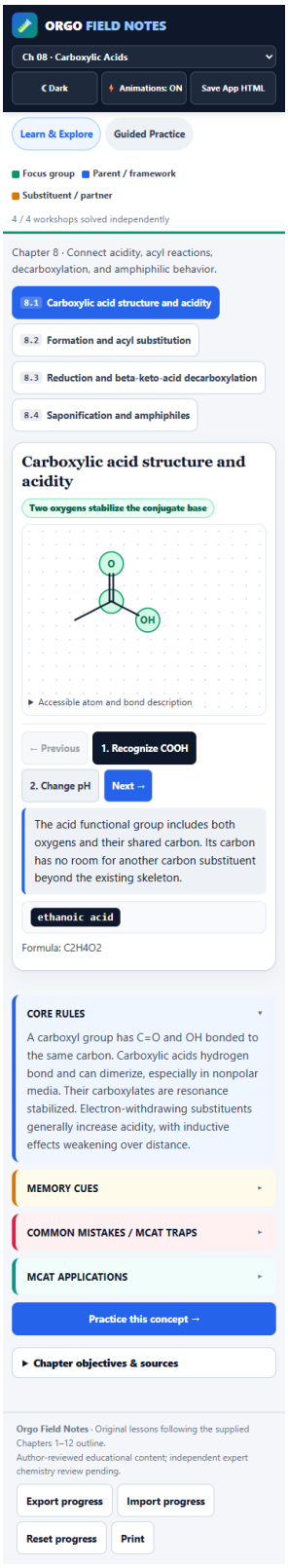
# Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

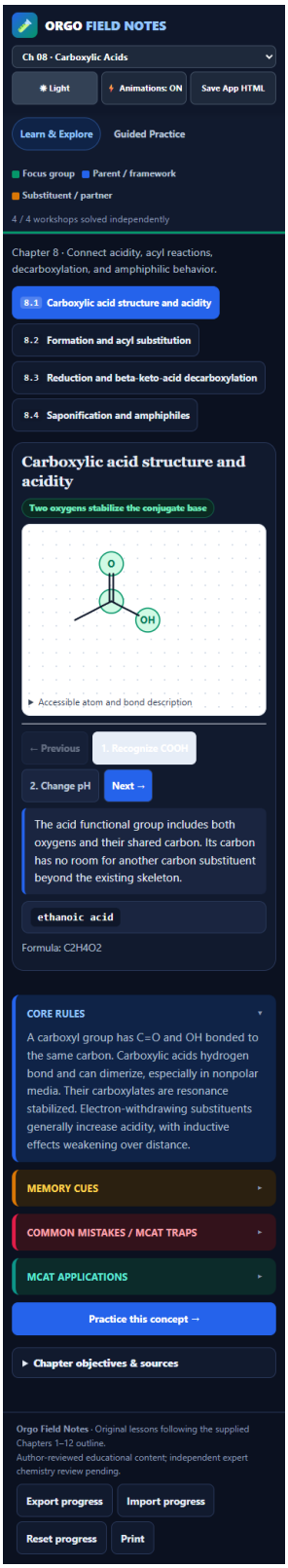
Screenshot: layout-desktop-light



Screenshot: layout-mobile-light



Screenshot: layout-mobile-dark




# Execution evidence

53 screenshots and 12 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.